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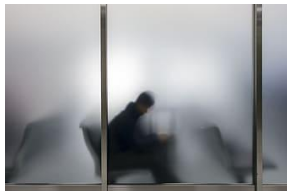
**LIGHT AND
SPACE**



2023

9P - 1a

Review of Light

			
Moon	Brick	Frosted glass	Light bulb
			
The Sun	Glass window	Milk	Mirror

1 Which one object above is transparent?

~~Glass window~~.....

2 Which objects above are translucent?

~~Frosted glass, Milk~~.....

3 Which objects above are opaque?

~~Brick, Moon~~.....

4 Which object is the best at reflecting light?

~~Mirror~~.....

5 Which object is the best at absorbing light?

~~Brick~~.....

Unit 1 - Reflection and Image in a Plane Mirror

6 Which object is seen because it reflects sunlight?.

~~The Moon~~.....

~~It can be seen because it reflects sunlight towards us; it is not a light source~~.....

7 Which object is a light source when it is switched on?

~~Light bulb~~.....

~~It is a light source (luminous) only when switched on~~.....

8 Is the sun an example of a luminous or non-luminous object? ~~luminous~~.....

Why is the sun luminous? : ~~It emits (gives out) its own light~~.....

9. Is the moon an example of a luminous or non-luminous object? ~~non-luminous~~.....

Explain: ~~The Moon does not emit its own light; it only reflects light from the Sun~~.....

10. Extension: Draw a diagram to show how shadows are cast on the surface of the Earth during a solar eclipse. Explain or label why this happens.

· Diagram: Sun, Moon and Earth in a straight line. The Moon blocks the Sun's light, so its shadow falls on part of the Earth's surface. Light travels in straight lines, so the shadow is a dark, sharp patch (total eclipse) surrounded by a lighter shadow.

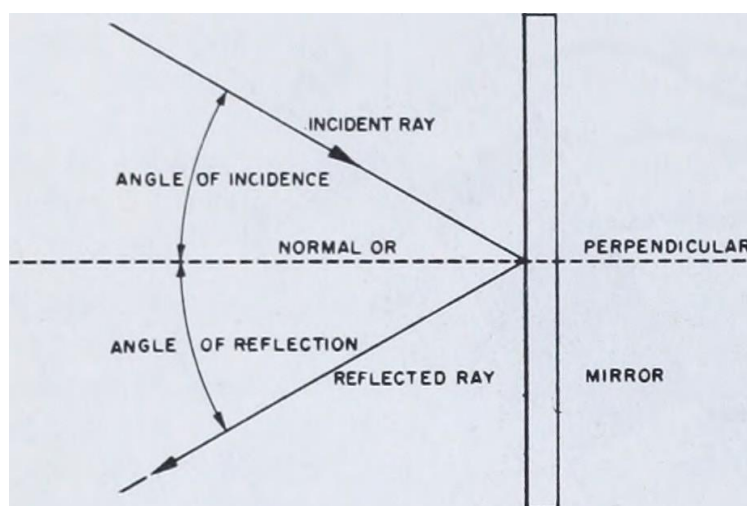
9P - 1b Investigating Reflection

Smooth or Rough?

Is your skin smooth or rough? It may seem smooth to your touch, but for light even apparently smooth surfaces are rough and SCATTER light.



You are going to find out if there is a relationship between the **angle of incidence** and the **angle of reflection** for a smooth surface like a plane mirror.



Prediction

I think the angle of reflection will be the same as (smaller than / greater than / the same as) the angle of reflection.

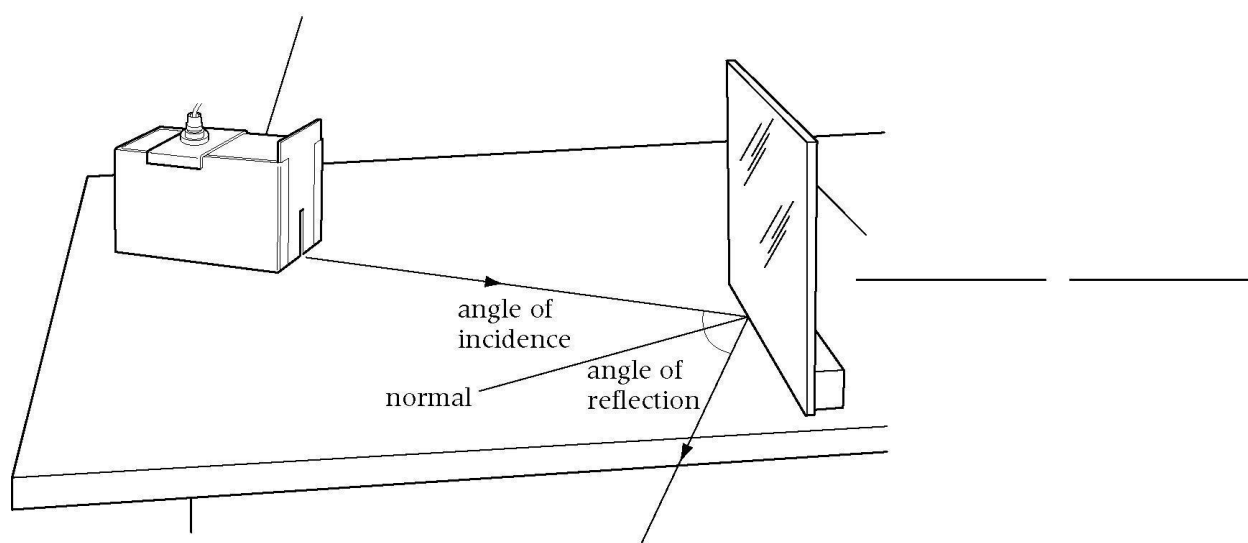
Unit 1 - Reflection and Image in a Plane Mirror

Label the diagram using words from the apparatus list.

Apparatus

- **Ray box**
- **Plane mirror**
- **Pencil**
- **Ruler**

Labels on the diagram: 'ray box' at the light source, 'plane mirror' at the reflecting surface, 'ruler' along the paper, 'pencil' marking the mirror position and ray paths.



Method

1. I will find out what the relationship is between the angle of incidence and the angle of reflection.
2. Using a piece of paper, I will mark down the position of the mirror.
3. I will use a protractor to measure a 90° line to the plane mirror. This line is called the normal and is drawn with a dotted line.
4. I will shine a ray of light onto the mirror using the ray box and label this the Incident Ray.
5. I will draw the position and label the reflected ray.
6. I will use a protractor to measure the Angle of Incidence and the Angle of Reflection, remembering that these are measured FROM THE NORMAL and not from the surface of the mirror.
7. I will record the two angles in the table.
8. I will then repeat this for 4 different angles of incidence.

Unit 1 - Reflection and Image in a Plane Mirror

Watch this video if you cannot do the experiment at home

https://www.youtube.com/watch?v=V0Z_nk5JS_E

Results

Angle of Incidence in ° (Angle at which light hits the mirror)	Angle of Reflection in ° (Angle that light is reflected at)

Conclusion

I noticed that the angle of incidence is the same as the angle of reflection.

This is the Law of Reflection

Extension: (Evaluation)

1. What were the main errors in this practical?

~~The ray of light was too wide, so it was hard to judge exactly where it hit the mirror.~~

~~Reading the protractor by eye was inaccurate and the lines drawn were too thick.~~

2. How could you have made this practical more accurate?

~~Repeat each measurement three times and take an average.~~

~~Use a sharp pencil and a thin ray, and measure with the protractor at eye level.~~

3. How could you use these principles to make a periscope? Find out how periscopes work and draw out a periscope design.

· Periscope: two plane mirrors set at 45 degrees, one at each end, facing each other. Light from the object reflects off the top mirror down the tube, then off the bottom mirror into the eye, so you see over or around an obstacle.

9C - 1c Where is the Image in a Plane Mirror?

- 1 Hold a plane mirror and look at yourself. Is the image that you see of yourself bigger, smaller or the same size as you?

~~The same size as me.....~~

~~The image is not magnified or reduced.....~~

- 2 Is the image that you see of yourself nearer to the mirror, further away or is it the same distance behind the mirror as you are in front of the mirror?

~~The same distance behind the mirror as I am in front of it.....~~

~~The image is the same distance behind the mirror as the object is in front.....~~

- 3 Now move away from the mirror. How does the image appear to move?

~~The image appears to move further away, staying the same distance behind the mirror as I am in front~~

~~The image and the object always move together.....~~

- 4 Now move towards the mirror. How does the image appear to move?

~~The image appears to move nearer, keeping the same distance behind the mirror as I am in front~~

~~The image moves towards me at the same speed I move towards the mirror.....~~

- 5 a) Write your name a piece of scrap paper. Use the mirror to look at the image of your name. Draw what it looks like in the space below.

~~My name is written back to front (reversed left to right) in the mirror.....~~

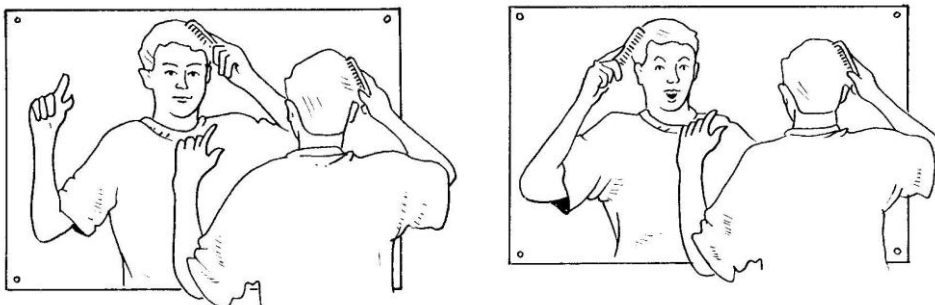
~~The letters are not upside down, but the order of the letters is reversed.....~~

- b) Is the image of your name upside down or the right way up? Is the image back to front? (Extension: Find out what this is called. (Clue; Lateral..... Inversion.....))

~~The right way up.....~~

~~Yes - the image is back to front.....~~

c) Which image is correct? Can you explain why?



4 Complete these sentences.

A flat mirror is called a plane (plane/plain) mirror. When a ray of light hits the mirror it is reflected (deflected/reflected). You see a reflection in the mirror called an image (image/picture). The image in the mirror is the same size as (bigger than/smaller than/the same size as) the object. The image appears to be the same distance from (nearer/further away from/the same distance from) the mirror as the object.

You cannot see an image (image/picture) using a rough surface like paper. This is because light rays (rays/streams) hitting the paper are scattered (scattered/absorbed) in all directions.

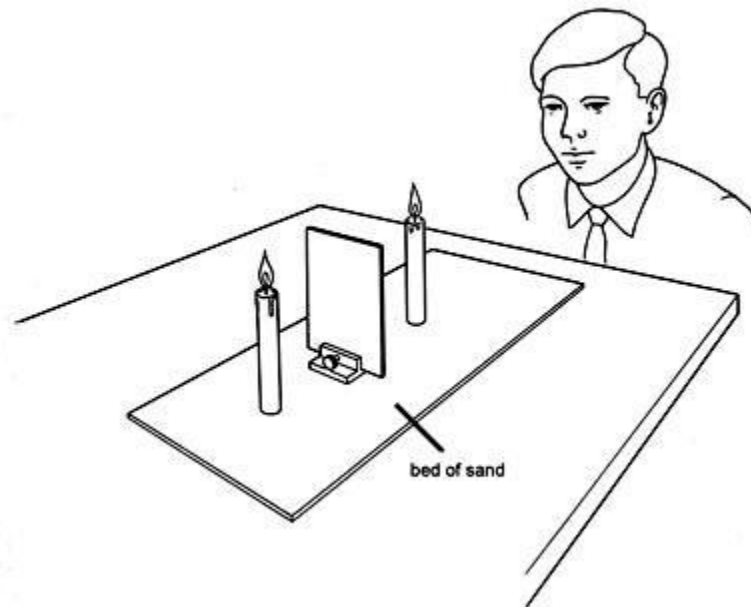
Extension: The image also seems to swap left and right.

This is called lateral inversion.

Where is the Image in a plane Mirror?

A sheet of plane glass can act as a 'two way mirror'.

Can you light one candle but make it look like both candles are lit?



How are the positions of the lit and unlit candles related?

~~They are the same distance from the glass, one in front and one behind.....~~

~~They lie on the same straight line through the glass, so the reflection appears to sit on the unlit candle.
Yes - put an unlit candle exactly where the reflection of the lit candle appears
behind the glass, and only the lit candle needs to be lit.~~

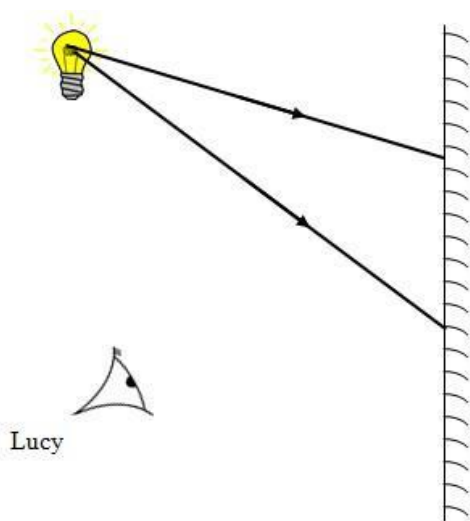
What happens when the lit candles moves closer and further from the glass?

~~The image moves the same distance behind the glass, keeping the same distance from the glass as the real candle.~~

~~The two candles therefore only line up when they are placed symmetrically about the glass..~~

Unit 1 - Reflection and Image in a Plane Mirror

Ray Diagram to find the Position of the Image in a Ray Diagram



1. Draw the reflected ray for both incident rays. Make sure the angles are correct, use your protractor and don't forget the normals!
2. Now extend the rays backwards behind the mirror using dotted lines (these are called 'construct rays'). This is where the light reaching Lucy appears to come from!
3. The image is where the dotted lines meet! Mark this as position I.
4. Now measure how far behind the mirror the image is compared to how far the light bulb is in front.

Conclusion

The image in a plane mirror is the same **distance**..... behind the mirror as the **object**..... is in front of the mirror.

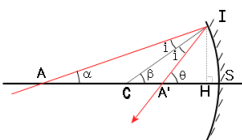
The image is **virtual**..... This means the light is not really there it just looks like it is there!

(We use the same word for “ **virtual**..... reality” in computer games. It looks real, but it is not real)

9P - 1d Ray Diagrams for Curved Mirrors

Use the ray boxes to give multiple rays and see how the curved mirrors reflect light.

Concave mirror



A concave mirror curves in (like a cave goes into a cliff)

Draw around the inside of your mirror 'hatch' the outside. Draw your 3 parallel incident rays and the reflected rays.

- Draw 3 parallel incident rays hitting the concave mirror. They reflect and cross over in front of the mirror, meeting at the focal point (they converge).

Conclusion

A concave mirror will cause parallel light to **converge**..... (converge / diverge)

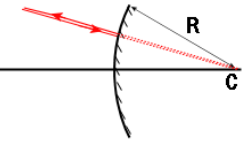
What are the uses of a concave mirror?

Reflecting telescopes (to collect light from distant objects).....

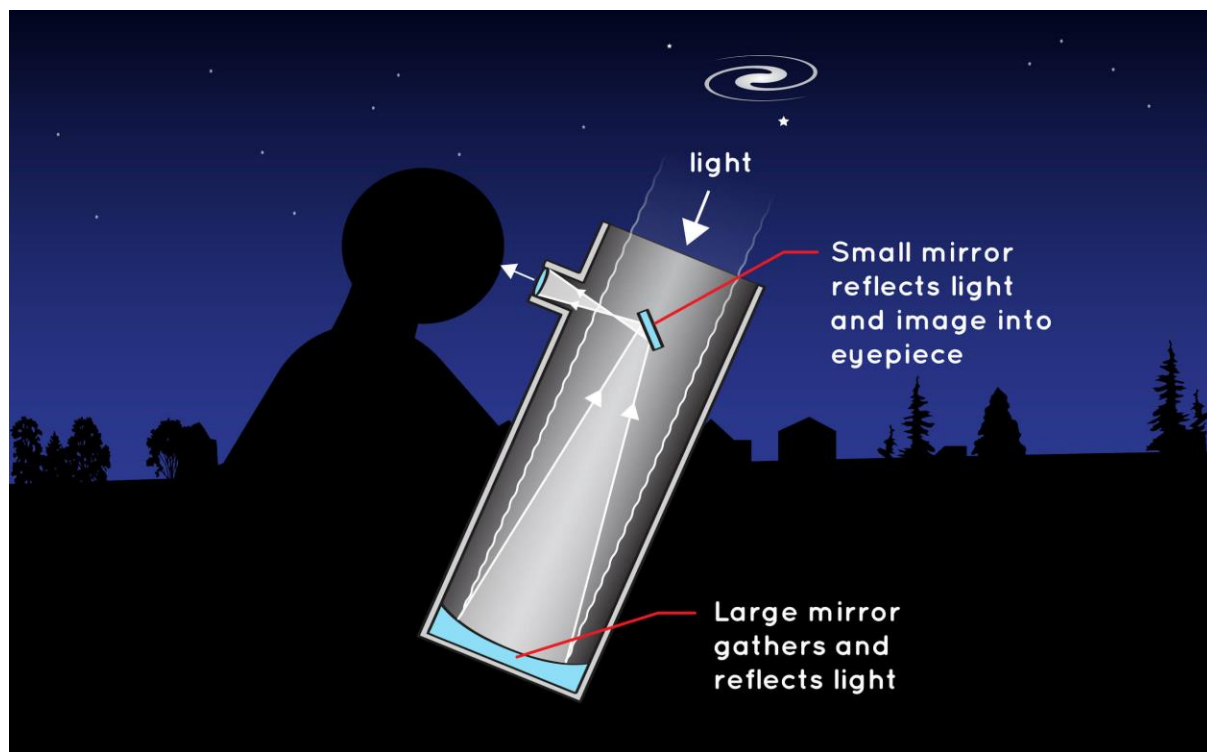
Shaving and make-up mirrors that magnify the face.....

Dentists' mirrors and headlight/spotlight reflectors.....

Convex mirror

 A convex mirror curves outwards	Draw around the outside of your convex mirror 'hatch' the inside. Draw your 3 parallel incident rays and the reflected rays. · Draw 3 parallel incident rays hitting the convex mirror. They reflect and spread out (diverge) as if they came from a focal point behind the mirror.
Conclusion A convex mirror will cause parallel light to diverge..... (converge / diverge) What are the uses of a convex mirror? Car wing mirrors and rear-view mirrors Shop and security mirrors, which give a wide view of the whole shop Road safety mirrors at blind corners and driveways	

The Reflecting Telescope.



Light from distant objects like the moon arrives **parallel**..... (parallel / at random angles)

A reflecting telescope uses a **concave**..... (convex/concave) mirror

to **converge**..... (converge / diverge) the light so that the small plane mirror can reflect it towards our eye.

9P - 1e How do we use mirrors?

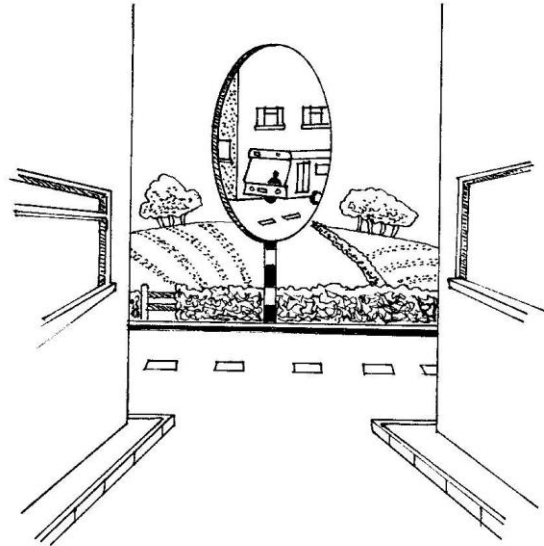


Why does the ambulance have the sign reversed?

Drivers see the ambulance in their rear-view mirror

A rear-view mirror (laterally inverts) the writing

so printing the sign reversed makes it read the right way round

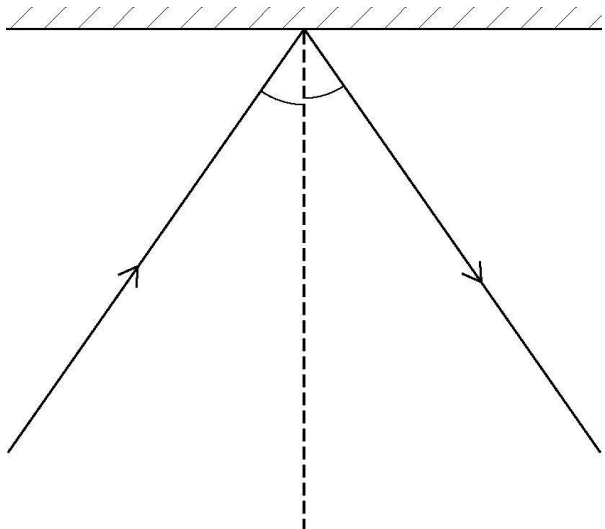
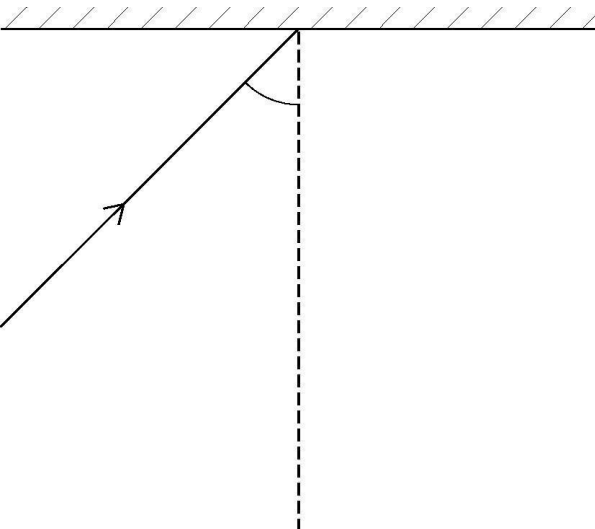


Why is the mirror helpful?

It lets the driver see the road and traffic behind

without turning round, which is safer.....

Unit 1 - Reflection and Image in a Plane Mirror

 a Use the words from the box to label this diagram. normal plane mirror angle of incidence angle of reflection b Measure the angles in the diagram using a protractor and write the answers down. Measure the angle of incidence of 40 degrees (measured from the normal) Angle of reflection = 40 degrees (measured from the normal) exactly the same angle and label it the angle of reflection. The angle of incidence equals the angle of reflection (Law of Reflection)	 Measure the angle of incidence on the diagram. Use a protractor to draw in the angle of reflection at the correct angle and label this angle on your diagram.
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Extension: Find out what is meant by a '**virtual**' image compared to a '**real**' image.

A real image is formed where light rays actually meet, so it can be caught on a screen.....

A virtual image is formed where rays only appear to come from $\therefore$ they do not really meet.....

A virtual image cannot be caught on a screen; it is formed behind a mirror or where a lens magnifies.

Real images are always inverted; virtual images are the right way up.

Is the image in a plane mirror virtual or real?

Virtual.....

The reflected rays do not actually meet behind the mirror,.....

they only appear to come from behind it (construct rays),.....

so the image cannot be caught on a screen,.....

Is the image in a camera virtual or real?

Real.....

The lens makes the light rays actually meet (converge) on the sensor,.....

so the image can be caught on a screen.....

The camera image is inverted and smaller than the object.

9P 2a Refraction

Introduction

All waves can refract.

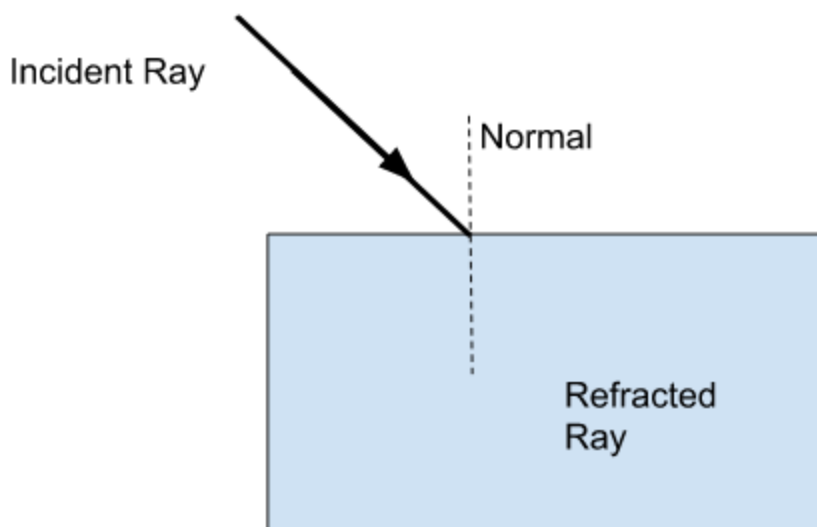
Refraction is caused by a change of **medium**.

When light changes its medium (*the material it is travelling through*) it **refracts** and changes **direction** because the light has changed its **speed**.



Experiment 1 - Investigating Refraction

1. Get a glass block and draw around it on paper. This can be in the space on the following page, graph paper or plain paper.
2. Draw in a normal, just to the left of the centre of the block as shown below.
3. Shine a ray of light into the glass block and label it the 'incident ray'. Add arrows to show where the ray refracts and emerges.



Emergent Ray

Experiment 1 - Space for drawing.

· Drawing: outline of the glass block with a normal drawn to the left of centre. The incident ray enters the block and bends towards the normal (refracts), travels through, then bends away from the normal as it emerges. The emergent ray is parallel to the incident ray.

Conclusion (circle the correct response in brackets)

• I noticed that the ray of light is refracted (*away from / towards*) the normal when it goes into a glass block.

towards

• The ray of light is refracted (*away from / towards*) the normal when it comes out of a glass block.

away from

• The ray coming out of the glass block is (*parallel / at right angles*) to the ray going into the block.

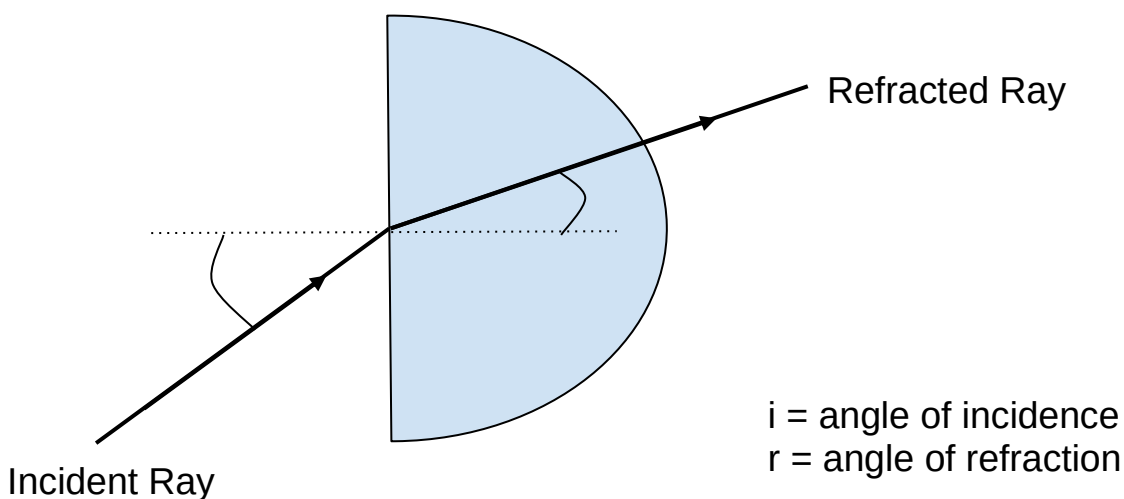
parallel

Experiment 2 - Investigating the relationship between the angle of incidence and the angle of refraction

Aim:

To establish a relationship between the **angle of incidence** and the **angle of refraction** for a semi-circular glass block.

1. Get a semi-circular glass block and draw around it on paper. This can be in the space on the following page, graph paper or plain paper.
2. (on the next page).
3. Draw the normal in the centre of the flat edge of the block as shown below.
4. Shine a ray of light into the glass block at 10° to the normal line.
5. Note where the refracted ray leaves the circular side of the glass block by making a mark with a pencil.
6. Draw the path of the refracted ray by connecting the points where it entered and exited the glass block.
7. Use a protractor to measure the angle of refraction.
8. Repeat steps 3-6 for angles of incidence of 20° , 30° , 40° , 50° , 60° , 70° and 80° .



Unit 2 - Refraction

Experiment 2 - Space for drawing.

Results

Angle of Incidence ($^{\circ}$)	Angle of Refraction ($^{\circ}$)
10 Angle of refraction = about 7 degrees	
20 Angle of refraction = about 13 degrees	
30 Angle of refraction = about 19 degrees	
40 Angle of refraction = about 25 degrees	
50 Angle of refraction = about 31 degrees	
60 Angle of refraction = about 35 degrees	
70 Angle of refraction = about 39 degrees	
80 Angle of refraction = about 41 degrees	

Unit 2 - Refraction

Data Processing

Draw an appropriate line graph of your results. Here are a few things to consider:

- Heading, labels with units on each axis
- Suitable scales that occupy minimum $\frac{3}{4}$ of each axis
- Plot all points accurately
- Draw a curve or straight line of best fit that ignores points that do not fit the pattern.
- Discuss trends or patterns observed in the graph.

· Graph: x-axis angle of incidence in degrees (0-80, using at least $\frac{3}{4}$ of the axis), y-axis angle of refraction in degrees (0-45). Plot all points and draw a smooth curve of best fit; the curve rises steeply at first then flattens.

Conclusion

(What is relationship between the angle of incidence and the angle of refraction?)

As the angle of incidence increases, the angle of refraction also increases

The angle of refraction is always smaller than the angle of incidence because light slows down and bends towards the

The graph is a curve, not a straight line through the origin, so the two angles are not directly proportional

The angle of refraction increases more and more slowly as the angle of incidence gets bigger

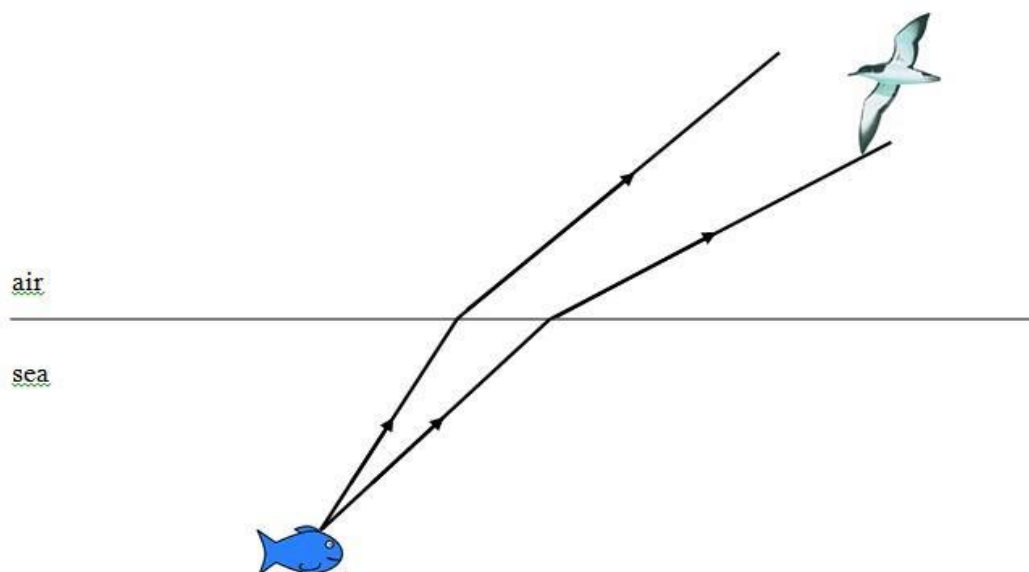
9P 2b Effects of Refraction

Apparent Depth

Watch the first 2min45s of following video on REAL and APPARENT DEPTH:
[Real and Apparent Depth](#)

<https://www.youtube.com/watch?v=KgZWf5WECg4>

1. A shearwater is looking for a fish to catch. Light from the fish refracts as it leaves the water.



From the shearwater's point of view where does the fish appear to be? Follow the rays backwards with dotted lines to where they meet, this is where the fish appears to be!

The distance between the surface of the water and where the rays meet (the image) is called the APPARENT DEPTH. The distance between where the object actually is and the surface of the water is called the REAL DEPTH.

The fish appears to be **shallower**..... (shallower / deeper) that it really is.

This is called APPARENT DEPTH. To catch the fish the shearwater would have to

dive **below**..... (above / below) where the fish appears to be.

· Extend the two emergent rays backwards into the water with dotted lines; they meet at a point above the real fish - this is the apparent position of the fish.

2.

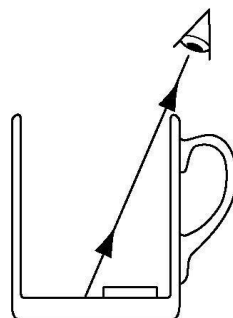


The goggles in the pool appear to be shallower (deeper / shallower) than they really are.

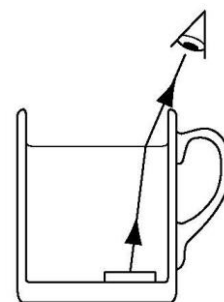
This is because light rays change direction (direction / curve) when they leave the water.

We call this refraction (reflection / refraction).

3. Coin in a cup.



coin is invisible



you can see the coin

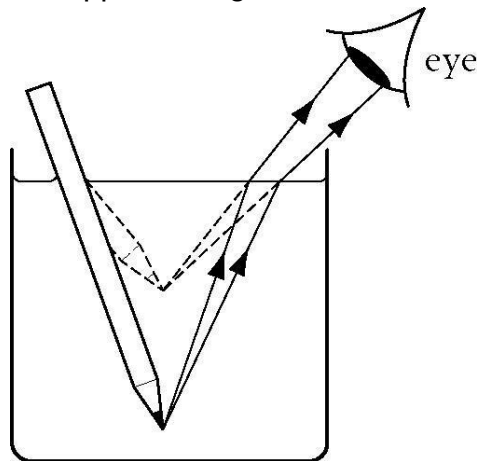
Explain how this trick works.

Light from the coin refracts (changes direction) as it leaves the water.....

The rays bend away from the normal, so your eye assumes they came in straight lines from a point higher up

The coin therefore appears to rise up and become visible over the rim of the cup.....

4. This diagram shows what happens to light which is reflected by the pencil.



a) Does the pencil seem to be deeper or nearer the surface of the water?

Nearer the surface.....

It looks shallower than it really is.....

Because light from the pencil refracts as it leaves the water.....

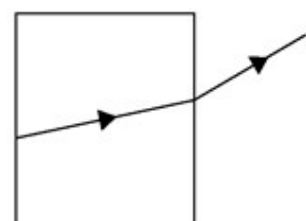
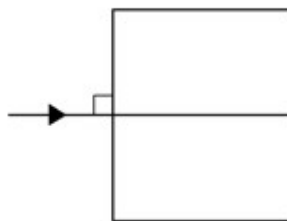
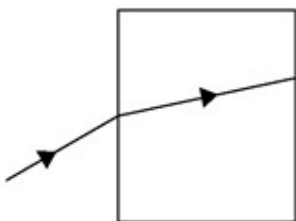
b) Explain why the pencil appears to bend.

Light from the part of the pencil under the water changes speed and refracts as it leaves the water.

The rays reaching your eye appear to come from a shallower position, so the light appears to bend

The underwater part of the pencil therefore looks displaced sideways, so the pencil appears bent at the surface

5. Add the missing ray to each glass block.



In each glass block, continue the ray bending towards the normal as it enters the glass, then bending away from the normal as it leaves, emerging parallel to the original incident ray.

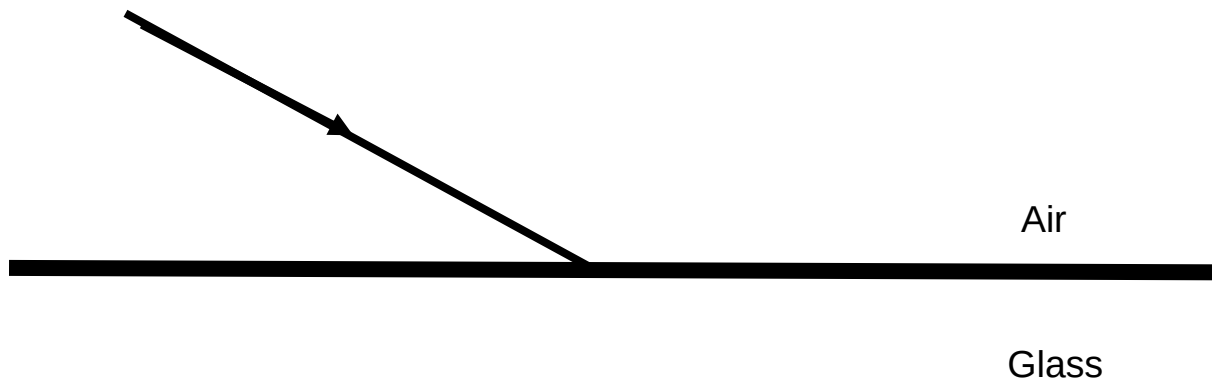
6. Draw a ray diagram below and include the following labels

· Ray diagram: incident ray in air meets the air-glass boundary; dotted normal drawn at 90 degrees to the surface; angle of incidence is the angle between the incident ray and the normal; the ray bends towards the normal in the glass and the angle of refraction is the angle between this ray and the normal.

a) The normal

b) The angle of incidence

c) The angle of refraction



Remember to draw the normal with a dotted or dashed line

7. a) Which is denser, glass or air? Glass is denser than air.

b) How does the density of the material affect the speed of which light travels through it The denser the material, the slower light travels through it, so the light bends more.

Extension:

8. What causes refraction of light? Try to explain in terms of the speed of the light. You can draw diagrams or use comparisons/analogies to help you.

Refraction is caused by light changing speed. When light passes from one medium into another of different density it speeds up or slows down, and this change of speed makes it change direction at the boundary.

9. Why do stars 'twinkle'?

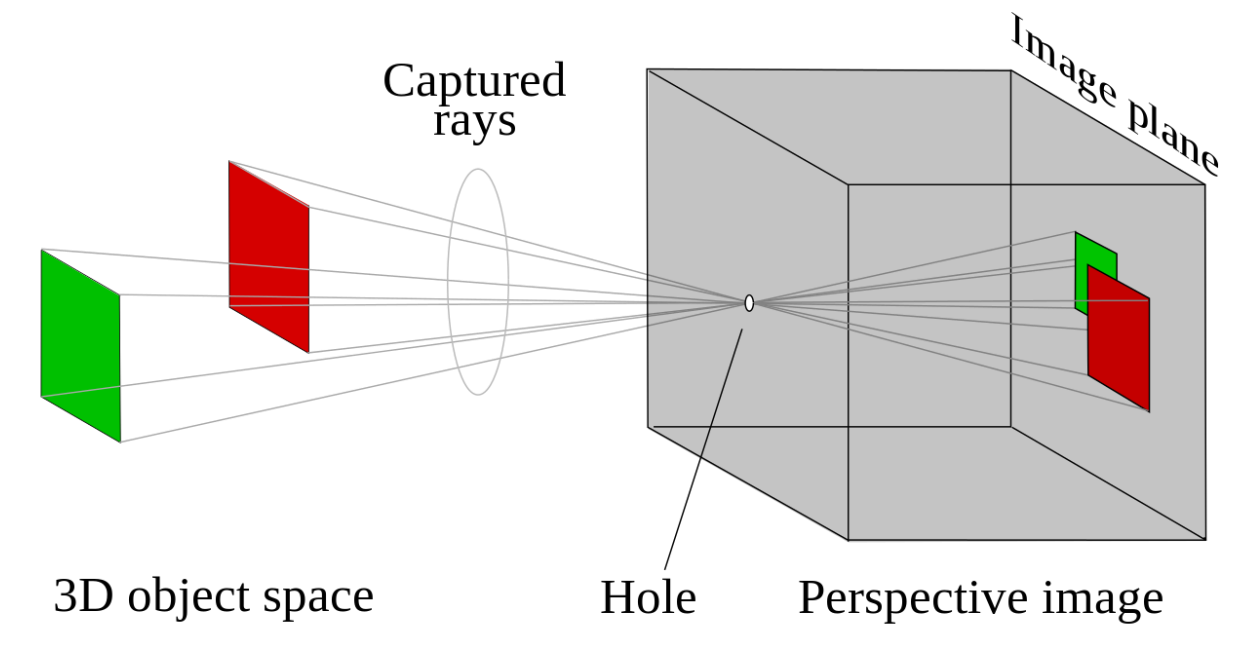
Starlight passes through layers of the Earth's atmosphere which have slightly different densities. The light is refracted by different amounts as the air moves, so the direction and brightness of the light reaching us keeps changing and the star appears to twinkle.

9P - 3a Investigating Pinhole Cameras

Apparatus • Pinhole camera • Strong light source (lamp) • Weak light source (Candle)	Safety Do not look directly at the Sun. It will damage your eyes.
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How does a pinhole camera work?

Light passes through the pinhole and an image is seen on the screen at the back

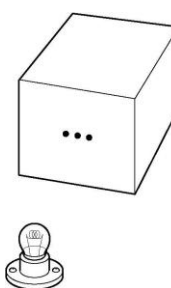
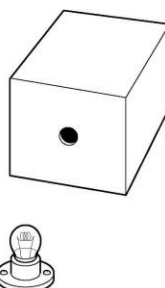
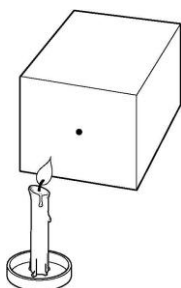
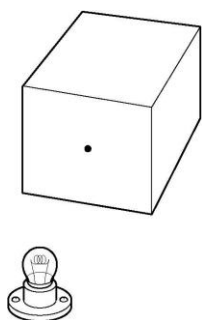


Unit 3 – Pinhole Camera – The Eye

Method

Carry out these investigations. Before you begin, discuss and predict what you think may happen and why.

Prediction: Prediction: a bright light source will give a bright, clear image; a dim light will give a faint image.	Prediction: Prediction: a bigger hole will give a brighter but blurrier image; a tiny hole will give a sharper but dimmer image.	Prediction: Prediction: each extra hole will make its own image, so we will see several overlapping inverted images.	Prediction:
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Prediction: the image will be upside down (inverted) in every case and smaller than the object.

Record your observations under each diagram

Observations: Observations: the bright light gave a bright, clear image; the dim candle gave a faint, dim image.	Observations: Observations: the larger hole gave a brighter but blurred image, and the very small hole gave a sharper but dimmer image.	Observations: Observations: with more holes we saw several separate overlapping images of the light source on the screen.	Observations: Observations: in every case the image was upside down (inverted) and smaller than the object.
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Conclusion

- The image in a pinhole camera was **Real**..... (Real / Virtual)
- The image in a pinhole camera was **inverted**..... (upright / inverted)
- The image in a pinhole camera was **diminished**... (magnified / diminished)

Questions

1 What is the difference between the images formed by a dim and a bright light?

A.bright light gives a bright, clear image, while a dim light gives a faint image.....

.The size, shape and position of the image are the same in both cases.....

2 What effect did changing the hole size have on the image?

A.larger hole gave a brighter but blurred image.....

A.smaller hole gave a sharper but dimmer image, because less light gets through.....

3 What happened when you changed the number of holes?

Each hole produced its own image on the screen.....

So more holes gave several overlapping inverted images of the object.....

4 Why does this experiment give evidence that light travels in straight lines?

.The image is inverted, which can only happen if the light from the top and bottom of the object crosses over at the hole

.This shows the light travelled in straight lines from the object, through the pinhole, to the screen

Unit 3 – Pinhole Camera – The Eye

Extension - Nature of Science

What is a 'camera obscura' and what was its influence on **art**?

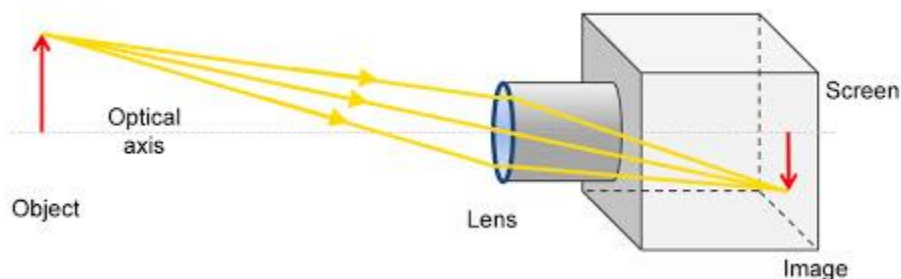
A camera obscura is a dark room or box with a tiny hole in one side.....
Light from outside passes through the hole and forms an upside-down image.....
of the scene on the opposite wall or screen, because light travels in.....
straight lines. Artists used it to trace accurate outlines and perspective, which influenced realist painting and early photography.

A Simple Lens Camera:

When the pinhole camera had a large hole the image was (**brighter** / dimmer) ,
but also became (sharper / **blurred**) .

Can a lens be used to focus the image and make it sharper?

Yes - a convex lens refracts (bends) the light rays so they converge and meet on the screen, giving a brighter, sharper (focussed) image.



Experiment with Lens Cameras

1. Try some of the different **convex lenses** available and see which one works best. Place the different lens in front of the aperture (hole). Does one create a focussed image?

Compare the length of the pinhole camera to the focal length of the lens which works best.

· Experiment: place each convex lens over the aperture in turn and look for a sharp image on the screen.

The image was sharpest (and the camera length needed) when the distance from the lens to the screen equalled the focal length of the lens.

Unit 3 – Pinhole Camera – The Eye

2. What can you do to help focus the image? Is this how real cameras focus?

~~Move the lens (or the screen) nearer to or further from the hole until~~
~~the image is sharp - changing the lens-to-screen distance changes the focus.~~
~~Yes - real cameras focus in the same way, by moving the lens.~~

3. Describe how the image appears using any appropriate words below

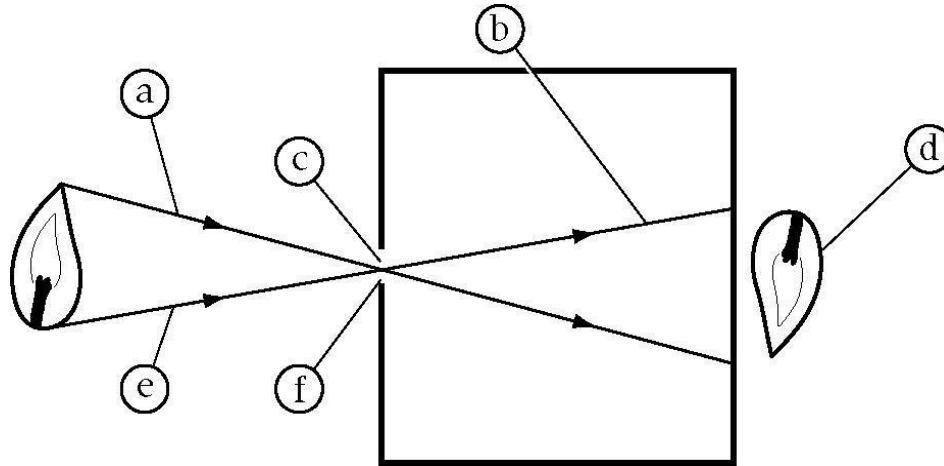
(do **not** use all the words):

(light, screen, image, upright, focussed, lens, inverted, left to right, real, virtual, enlarged, diminished, magnified)

~~The image is real and inverted - it is upside down and also~~
~~reversed left to right. It is focussed sharply on the screen~~
~~and it is diminished (smaller than the object). It is formed~~
~~because light travels in straight lines from the object,~~
~~through the lens, onto the screen,~~
~~backwards and forwards so the image falls exactly on the film or sensor.~~

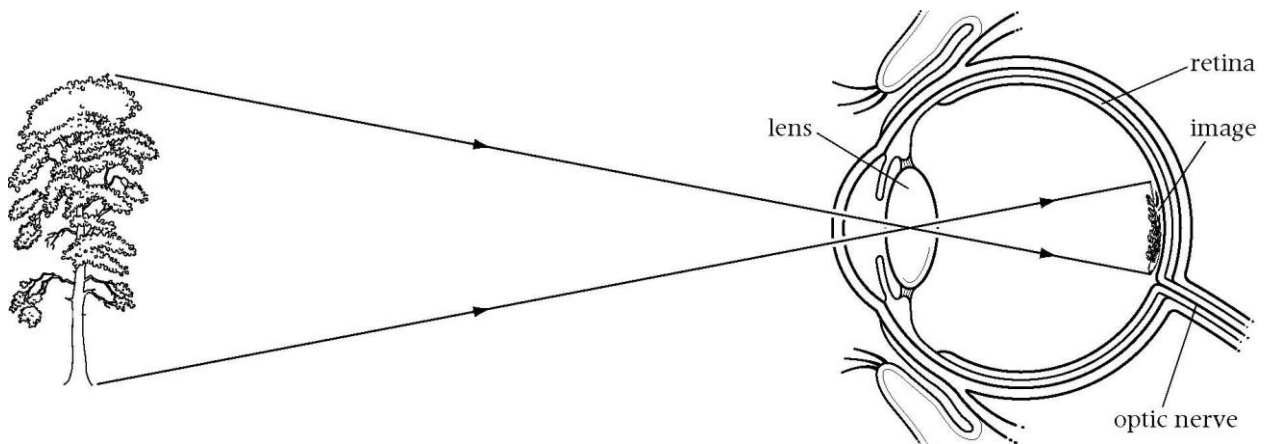
9P - 3b How Pinhole Cameras and Eyes work

1. Label the following pinhole camera using some of the statements below.



Statement

- small hole
- upside down image
- Light from one part of the flame can only reach one place on the screen.
- light ray from the bottom of the candle flame
- light ray from the top of the candle flame



2 Why is the image upside down?

Light travels in straight lines. A ray from the top of the object goes

through the pinhole and lands at the bottom of the screen (and the ray from the bottom lands at the top), so the image is inverted.

3 Why does a small hole give a sharp but dim image?

A small hole only lets a narrow beam of light from each point of the object through, so each point of the object forms just one point on the screen and the image is sharp. But very little light gets through a small hole, so the image is dim.

4 Give two ways in which a pinhole camera is similar to the eye.

Both have a small opening that lets light in. (pinhole / pupil) and both form a real, inverted image on a screen / the retina because light travels in straight lines.

5 Where in the eye is the image formed?

On the retina, at the very back of the eye (the light-sensitive layer of cells).

6 Which part of the eye sends messages to the brain?

The optic nerve, which carries the electrical impulses from the retina to the brain.

7 Why do we see an image the right way up?

The image on the retina is inverted, but the brain interprets and processes the nerve signals, so we see the object the right way up.

8 What does the lens do?

The lens refracts (bends) the light rays so that they converge
and focus onto the retina, forming a sharp, bright image.....

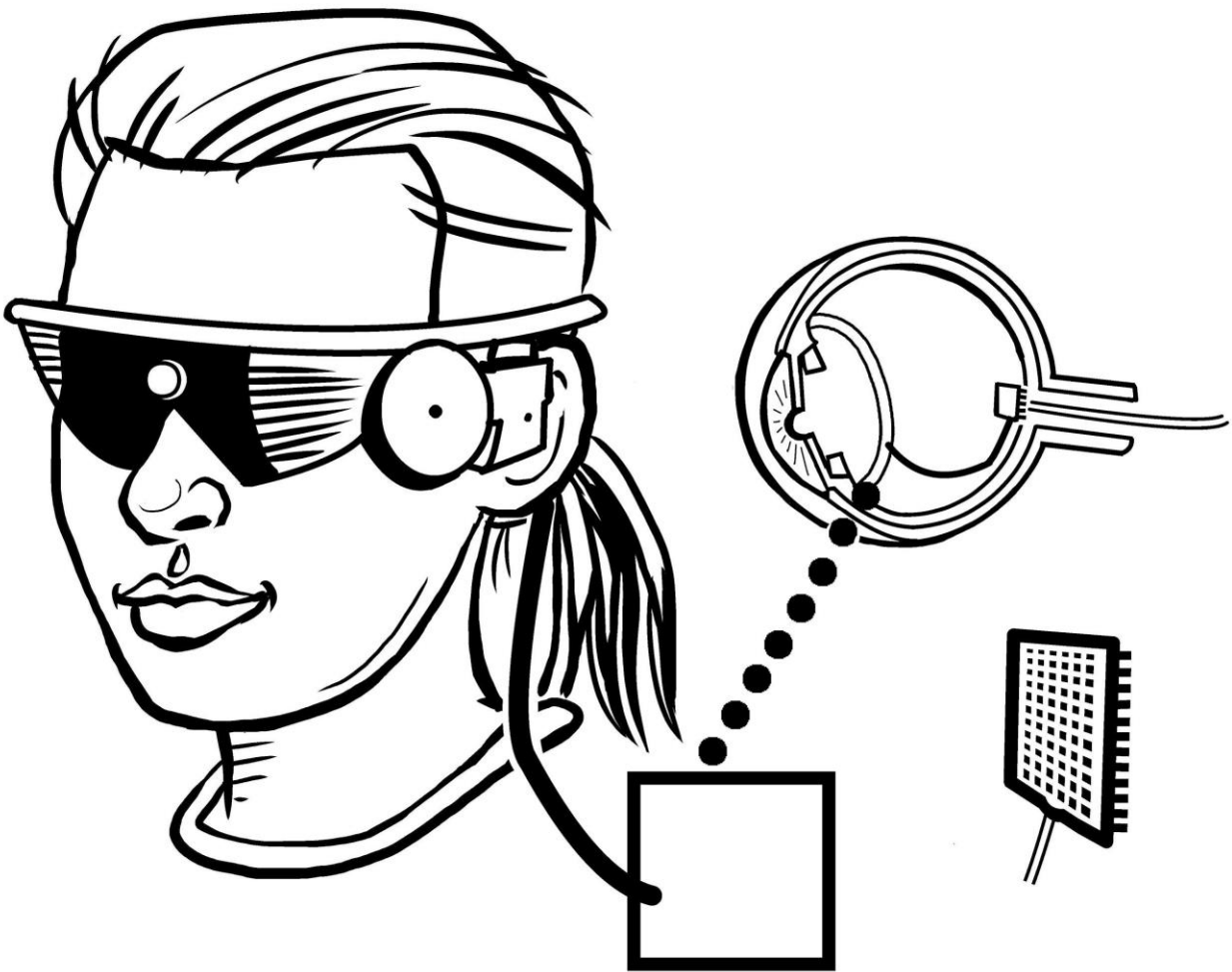
Extension:

9 How does the lens change in our eye to allow us to see things that are close or far away?

The lens changes shape - this is called accommodation. The ciliary
muscles make the lens thicker (more curved, shorter focal length)
to focus light from close objects, and thinner (flatter, longer
focal length) to focus light from distant objects.....

9P - 3c I Spy with my Bionic Eye

After exploring what we mean by light and how light rays behave you will find out how eyes and cameras work. You will then look at how science is improving the sight of people with poor vision using bionic implants.

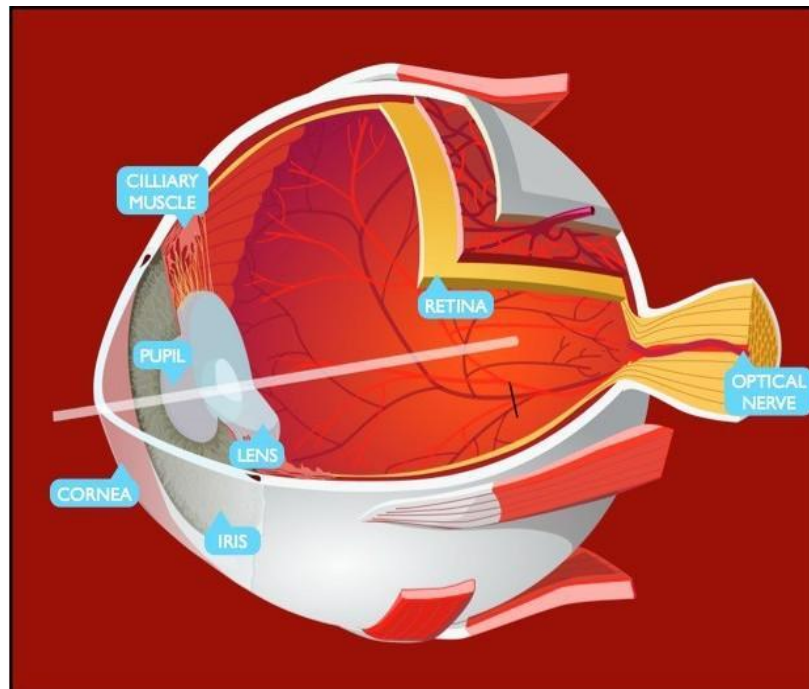


You will produce a leaflet or model or ICT presentation of a bionic eye implant system that improves vision for those with severe eyesight problems.

In your presentation, try and include some of the following:

- Describe what light is and some of its key properties.
- Explain how lenses work.
- Detail how the human eye functions, allowing us to 'see' objects.
- Detail how cameras work.
- Compare eyes and cameras, highlighting similarities between them.
- Explain how bionic eye implants function and lead to improvement in human vision.

9P - 3d Your eye as a camera

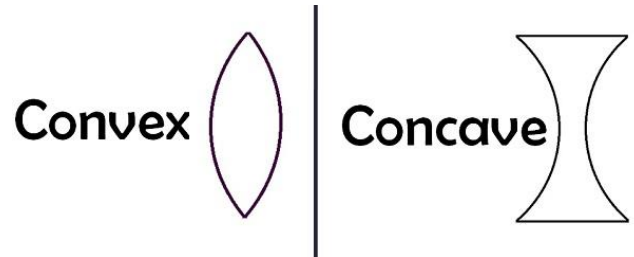


1. The **Cornea** and the **lens**, which is behind the iris, they are the eye's focusing elements. The cornea takes widely diverging rays of light and bends them through the pupil, the round opening in the central portion of the coloured iris.
2. The **Iris** adjusts the size of the **pupil** which acts like the aperture of a camera. The iris is a muscle which, when contracted, covers all but a small central portion of the lens, allows adjustable control of the quantity of light entering the eye so that the eye can work well in a wide range of viewing conditions, from dim to very bright light.
3. The **Retina** is the sensory layer that lines the very back of our eyes. It acts very much like the imaging sensor in a digital camera. The retina has numerous photoreceptor nerve cells that help change the light rays into electrical impulses and send them through the optic nerve to the brain where an image (of what we see) is finally received and perceived.

9P 4a - Lenses

Introduction

- Lenses are shaped pieces of glass which cause the rays of light passing through to change direction.
- There are two main types of lenses:



Experiment - Convex and Concave Lenses

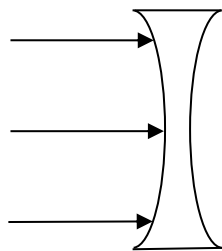
Aim:

To investigate how rays of light travel through convex and concave lenses.

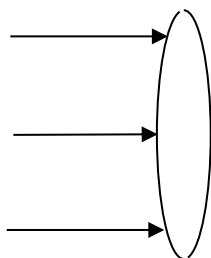
Prediction:

On the diagrams below, sketch what you think will happen as each of the incident rays pass through convex and concave lenses. Remember to use a ruler.

Hint: Think about what you know about refraction



Concave Lens



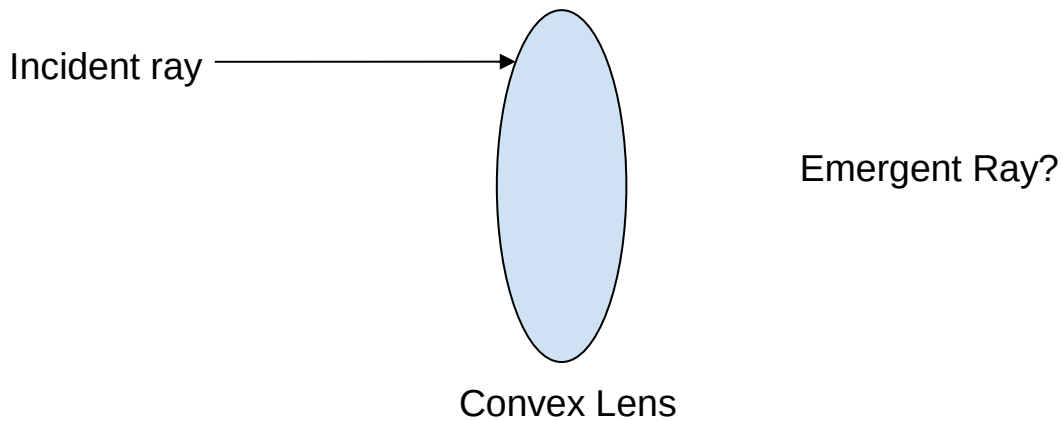
Convex Lens

• Concave (diverging) lens: the parallel incident rays should be drawn refracting outwards and spreading apart, diverging as if they came from a virtual focus on the same side as the incoming rays.

• Convex (converging) lens: the parallel incident rays should be drawn refracting inwards so that they converge and cross at the principal focus on the far side of the lens.

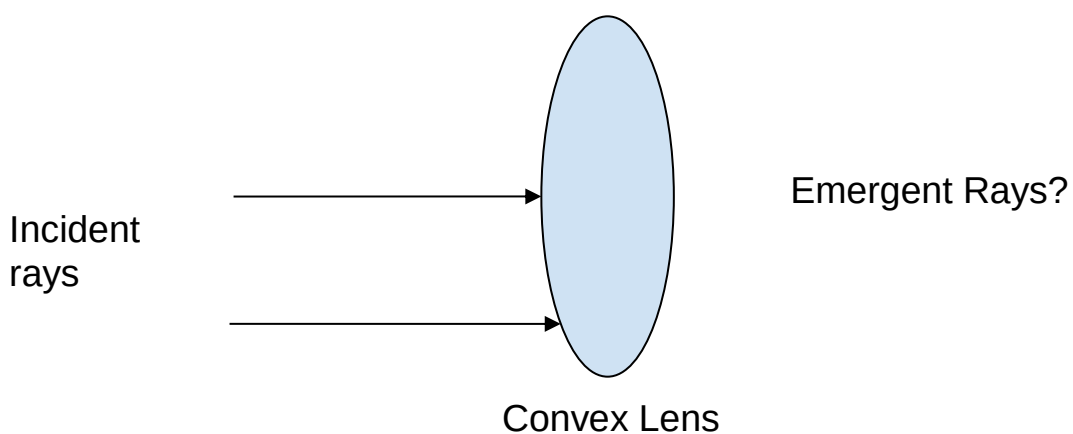
Method:

1. Get a convex lens shaped block and draw around it on paper (on the next page)
2. Add a line as shown below to indicate the incident ray. Note that it is positioned near the top of the lens.



· Emergent ray: the ray bends towards the normal as it enters the glass and away from the normal as it leaves, so it is refracted (bent) and converges towards the axis.

3. Align your ray box so that the ray of light is perfectly aligned with the incident ray line.
4. Sketch the path of the emergent ray
5. Repeat steps 2 - 4 for light entering at the middle of the lens and at the bottom as shown in the diagram below.

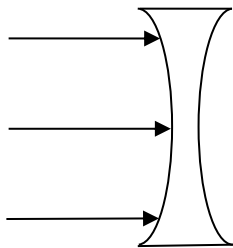


· Emergent rays: rays entering the top, middle and bottom of the convex lens all bend inwards so they converge towards the principal focus on the other side.

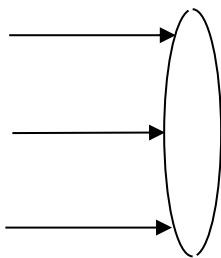
6. Repeat steps 1-5 for a concave shaped glass block.

Conclusion:

Complete the diagrams below by showing the path of the rays as they pass through and emerge from the concave and convex lenses.



Concave Lens



Convex Lens

· Concave lens diagram: draw the incident parallel rays diverging (spreading out) as they leave the lens, as if they came from a virtual focus on the same side.

Complete the conclusion statements below:

· Convex lens diagram: draw the incident parallel rays converging to the principal focus on the other side of the lens.

- A ray of light can change direction as it passes through a lens. This is caused by refraction. This is when a wave will change speed and sometimes direction as it goes from one medium into another.
- A convex lens is also known as a converging lens. The rays meet at a point called the focus.
- A concave lens is also known as a diverging lens.

converging	diverging	focus	refraction
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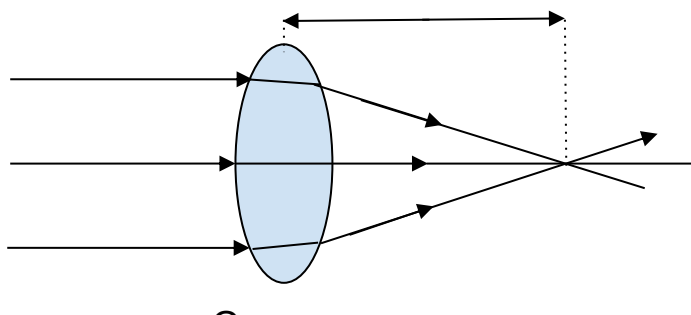
Unit 4 - Lenses

Extension:

Try different thicknesses of glass blocks.

9P 4b Investigating Focal Length

Experiment: Investigating the Focal Length of a Convex Lens:



Aim:

Your aim is to find the focal length of different convex lenses.

Method:

Your teacher will demonstrate how to measure the focal length of a convex lens.

There is also a video on ISLE.

Complete a step-by-step description of the method in the space below:

- 1.
 - 2.
 - 3.
 - 4.
 - 5.
 - 6.
1. Choose a convex lens and point it at a distant light source (e.g. a window or lamp).
 2. Move a white screen backwards and forwards behind the lens until a sharp image of the distant object is formed.
 3. Measure the distance from the middle of the lens to the screen - this is the focal length.
 4. Record the focal length in cm in a results table next to the lens colour code.
 5. Repeat the measurement three times and calculate an average to reduce random error.
 6. Repeat steps 1-5 for each of the other convex lenses (different colours/thicknesses).

Results:

<u>Lens (Colour Code)</u>	<u>Focal Length (cm)</u>	<u>Power (D) = $1/f$</u>
Record the colour code of each lens used, its focal length in cm and its power in dioptres (D). Example: red lens $f = 20\text{ cm}$ -> power = $1/0.20 = 5\text{ D}$; blue lens $f = 10\text{ cm}$ -> power = $1/0.10 = 10\text{ D}$. A longer focal length gives a lower power.		

Observations:

What do you notice about the orientation of the image??
Can you explain it?

The image is inverted - upside down - and also
reversed left to right. It is smaller than the object
and it is real, because it can be seen on the screen. This is because
light travels in straight lines, so the rays cross over at the lens and swap top/bottom and left/right.

Extension Activities:

1. Calculate the power of each lens above by using the the equation:

power = $1 / \text{focal length}$

power is measured in dioptres (D)
focal length must be converted into metres (m) for the equation to work.
power (D) = $1 / \text{focal length (m)}$. e.g. $f = 25\text{ cm} = 0.25\text{ m}$, power = $1/0.25 = 4\text{ D}$.

2. Draw a line graph of power v focal length.
3. Investigate how you would use a convex lens to magnify an image.
Hold the convex lens close to the object (nearer than the focal length) and look through it: the object appears enlarged. Move it to change the magnification.

9P 4c Applications of Lenses and the Refracting Telescope

Applications of Lenses



Uses of Lenses

Describe and explain some applications of lenses in the space below:

Lenses are used in cameras, spectacles, magnifying glasses, microscopes, telescopes and projectors; a

convex lens converges light to form a focused or magnified image, a concave lens spreads it out.

.

Magnifying glass -- a convex lens held close to an object forms a magnified, virtual image, so small print is easier to read.

.

.

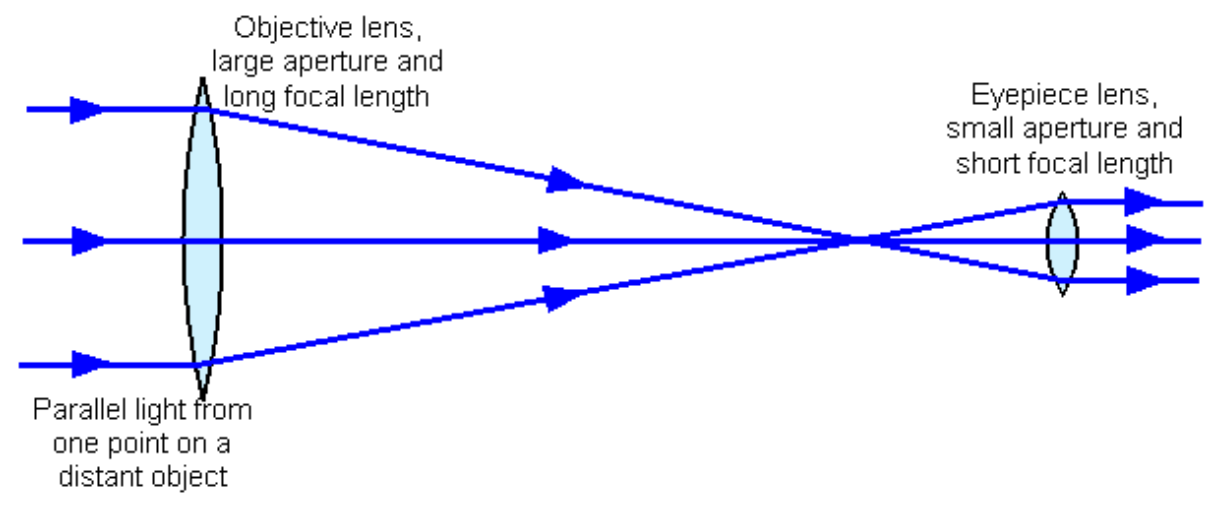
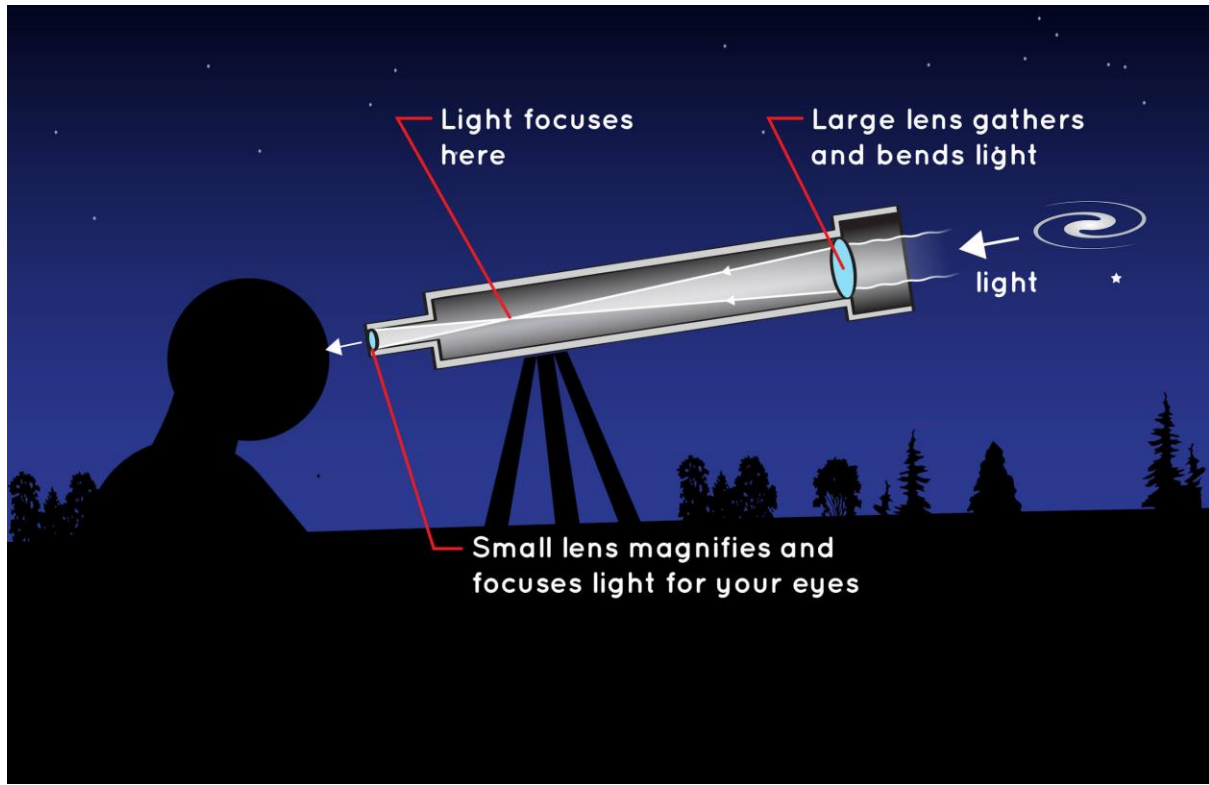
.

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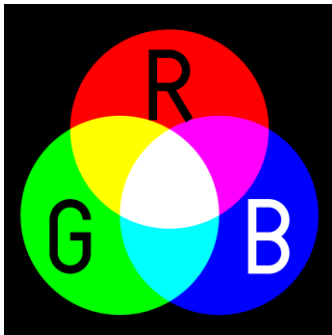
Cameras and the eye use a convex lens to focus light onto a sensor/retina; spectacles use converging lenses for long sight and diverging lenses for short sight; telescopes and microscopes use two lenses to magnify.

Refracting Telescopes:

A refracting telescope uses two lenses.



9P 5a - Colour Mixing



Class Experiment

Log onto: <https://www.focuselearning.co.uk/>

Username: **student@islandschool37651**

Password: **2y5wzdqhx**

Click on button named “ Essential Physics KS3”



Unit 5 - Polarization and Colour Mixing



Click on the Light section then Color and on the left-hand side panel

Start the experiment (only mix the light. **Do not mix the paint**)

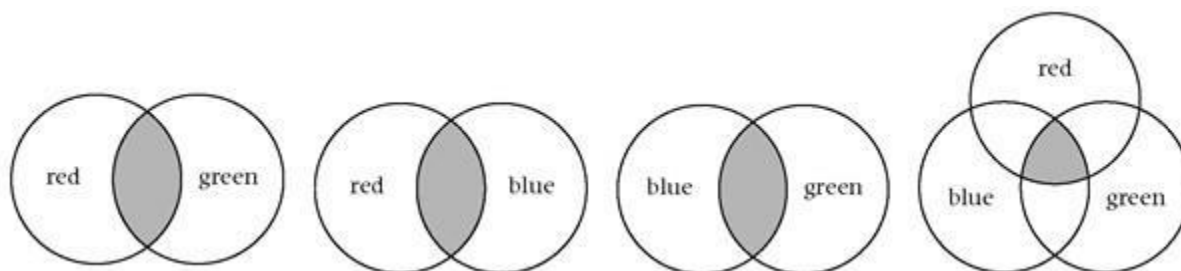
The interface shows a sidebar on the left with a menu similar to the first image, but with 'Colour and vision' selected. The main area is titled 'Colour and vision' and contains a Venn diagram with three overlapping circles: red (top-left), green (top-right), and blue (bottom). The intersections show yellow (red+green), cyan (green+blue), and magenta (red+blue), with white in the center where all three overlap. To the right of the diagram are three sliders, each labeled '100 %' and set to the maximum: 'Red 100 %', 'Green 100 %', and 'Blue 100 %'. Below the diagram are three buttons: 'Mixing light' (highlighted), 'Mixing paints', and 'Colour vision'. At the bottom, a text box says: 'All the colours of the rainbow can be made by mixing different amounts of red, green and blue light. These are called the primary colours.'

Full link: <https://www.focuselearning.co.uk/programmes/?frompack=&programme=essential-science-2018&page=light/light-colour&hl=colour%20mixing>

Unit 5 - Polarization and Colour Mixing

1 Write down the colour which is made when these lights overlap. Some colours are listed below to help you.

pink cyan magenta orange white yellow violet



yellow cyan magenta white

2 All the different colours that we can see in a rainbow are called the Spectrum of Visible Colours.

Three of these colours in the visible spectrum are primary colours.

The primary colours are Red , Green and Blue .

Primary colours are made from a single colour and not a combination of colours.

Two primary colours mixed together can make other colours

which are called secondary colours.

Red	secondary	spectrum	Green	primary	Blue
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Further Questions

In white light coloured objects reflect the light of their colour. The rest is absorbed.

Black objects absorb all colours. White objects reflect all colours.

If you have a red coat on, it appears red because only red light is reflected.

1. Which colours are absorbed by the red coat?

All the colours except red - orange, yellow, green, blue and violet are absorbed; only red light is reflected.

2. Which colours are reflected from:

a. a green T-shirt

b. black trousers

c. a white shirt

d. a blue jumper.

green.....

none (black absorbs all colours)

all colours (white)

blue.....

Black trousers reflect no colours - they absorb all colours of light, which is why they look black.

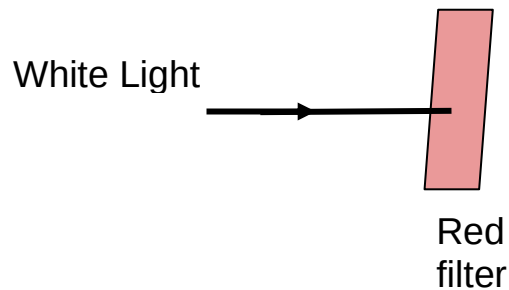
3. Fill in the colours:

Colour of object in white light	Colour of object in this coloured light			
	White light	Red light	Blue light	Green light
white <i>white object: in white light = white; in red light = red; in blue light = blue; in green light = green</i>				
red <i>red object: in white light = red; in red light = red; in blue light = black; in green light = black</i>				
blue <i>blue object: in white light = blue; in red light = black; in blue light = blue; in green light = black</i>				
green <i>green object: in white light = green; in red light = black; in blue light = black; in green light = green</i>				

Extension Questions

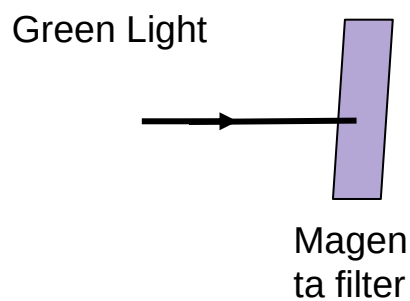
4. You can change the colour of light by using filters. Filters only allow light of the colour of the filter through and absorb all the other colours. Explain what light will get through:

a. White light hitting a red filter. **Red light only** - a red filter transmits red light and absorbs all the other colours.



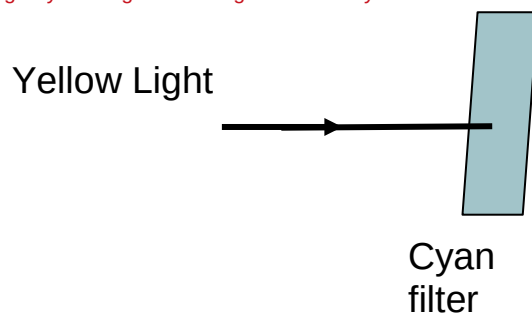
b. Green light hitting a magenta filter

No light gets through (the result is black) - a magenta filter transmits only red and blue, so it absorbs the green light.



c. Yellow light hitting a cyan filter

Green light gets through - yellow light is red + green and a cyan filter transmits blue + green, so only the green passes.



5. Which colours are reflected and absorbed by a yellow hat?

A yellow hat reflects yellow light and absorbs all the other colours (red, orange, green, blue and violet).

6. Which colours are reflected and absorbed by magenta trousers?

.....

7. Coloured Objects in different lights.

Magenta trousers reflect red and blue light (which together look magenta) and absorb all the other colours, including

Colour of object in white light	Colour of object in this coloured light			
	Blue light	Red light	Magenta light	Yellow light
yellow	yellow object: in blue light = black; in red light = red; in magenta light = red; in yellow light = yellow			
magenta	magenta object: in blue light = blue; in red light = red; in magenta light = magenta; in yellow light = red			
cyan	cyan object: in blue light = blue; in red light = black; in magenta light = blue; in yellow light = green			
green	green object: in blue light = black; in red light = black; in magenta light = black; in yellow light = green			

9P - 5b Investigating Polarizers

What are polarizers used for?

A polarizer may look just like a grey filter. However polarizers are key parts of Liquid Crystal Displays (LCDs).

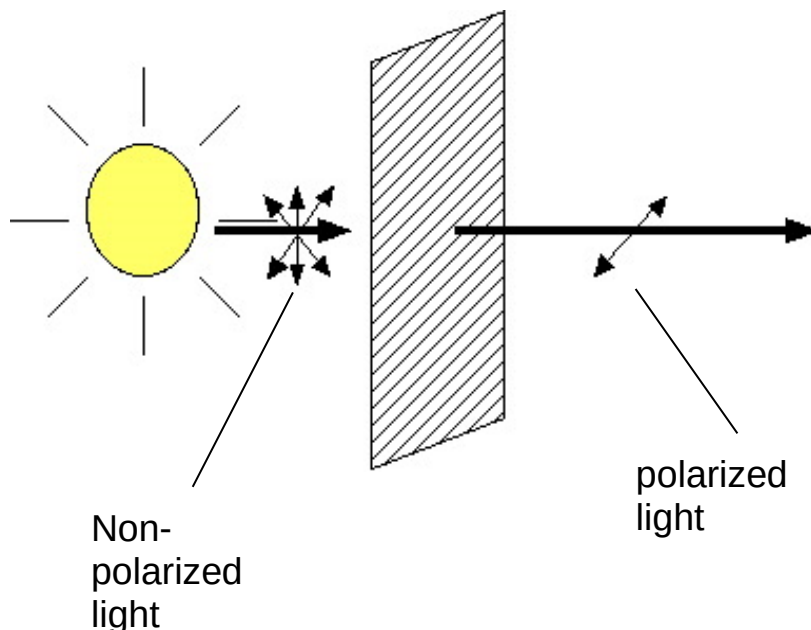
Try looking at a:

- laptop screen
- calculator screen
- Smart-phone screen
- Smart-watch display



Polarizers are used to reduce glare (polarizing sunglasses, camera filters) and in LCD screens, by letting through only light vibrating in one direction.

What is polarization?



Polarization is when light waves are made to vibrate in one direction (one plane) only. Non-polarized light from the Sun or a lamp vibrates in all directions; after a polarizer the light vibrates in one direction - the same direction as the polarizer.

Light from the sun or from a lamp is **non-polarized**, which means that the light wave vibrates in all directions.

After going through a polarizer the light is polarized meaning that the wave only vibrates in one direction. This direction of the polarization of the light is the same direction as the polarizer.

Can we find any polarizers or polarized light around us now?

Video on ISLE Lesson 5 - Polarizers all around.

<https://isle.island.edu.hk/course/view.php?id=1599>

Your teacher might show you what a polarizer does to the light from the LCD Projector.

What happens?

The polarizer at different angles will allow different amounts of light through.

In one direction the light was bright..... and 90o later

the light was dark.....

Magenta is a combination of red..... and blue.....

So since we have green and magenta light that means that all

three primary colours are there and we see white..... Light.

Take in a Movie

Can you think of somewhere you might be sitting and wearing polarizing glasses?

In a cinema the two polarizing lenses absorb different parts of the light from the screen and let different images through to the eyes. The brain puts these two images together and this creates the 3D..... effect.



Video on youtube: [Removing Glare from a water surface](https://www.youtube.com/watch?v=ChQgV7DlkQo)

<https://www.youtube.com/watch?v=ChQgV7DlkQo>

In a cinema when watching a 3D film - the polarizing lenses in the 3D glasses let each eye see only one of the two projected images.

Polarization in Nature

Light reflected from a water surface is naturally polarized. So using a polarizing filter helps cut out the 'glare'. Glare is unwanted light, just like noise is unwanted sound.

So polarizing filters are very useful in photography.

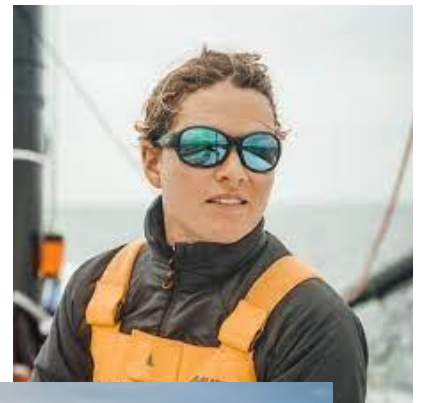
You can cut out the glare off the water and this helps see the light coming from the water.



'Eye Eye Sailor'

Why would a sailor wear polarizing sunglasses?

-
-
-
-



Summary

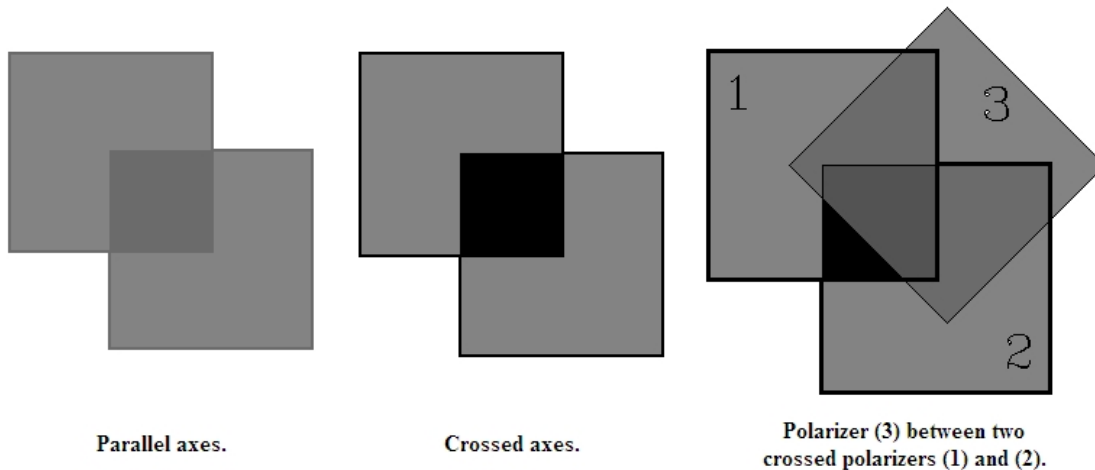
Polarizers can be used to absorb or transmit different light (more detail in the extension at the end), depending on the arrangement which we call the polarization of the light.

Light reflected from the water surface is naturally polarized, and polarizing sunglasses cut out this 'glare', so the sailor is not dazzled by the bright reflection. This lets them see through the water surface more clearly, e.g. to spot rocks, buoys or other boats, and it reduces eye strain in bright sunlight.

Investigating Crossed Polarisers

Download a light meter app onto your phone, or use a school light meter. Choose one that reads a brightness such as Lux and not an F number which is for photography.

What happens when you 'cross' two polarizers?



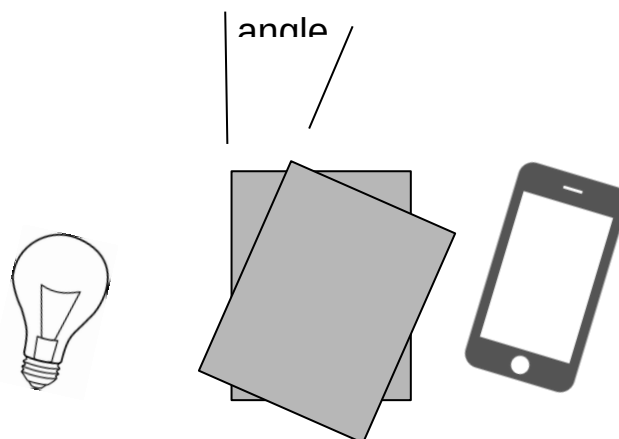
- A Polarizer orientates the light into one direction (this is polarized light).
- A second polarizer in the same direction (parallel) will let the light through
- A second polarizer 90° (crossed) will cut out all the light.

Research Question

How does the angle between two Polarizers affect the intensity of the light that passes through.

Apparatus

- Desk Light
- 2 polarizing filters
- Protractor
- Light meter (can be an app on your mobile phone)



No light passes through - the second polarizer blocks the light because its direction is at 90° to the polarized light, so the view goes black.

Method

1. Arrange a light source and two parallel polarising filters so that maximum light reaches your light meter.
2. Measure the light intensity (brightness) at angles of 10° , 20° , 30° , 40° , 50° , 60° , 70° , 80° and 90° between the 2 polarizers.
3. Repeat and calculate an average.
4. Draw a graph of your results.

Results

(Video on ISLE has some results you can use - Lesson 5)
<https://isle.island.edu.hk/course/view.php?id=1599>

Angle between polarizers (degrees)	Brightness of light meter. Repeat 1. (lux)	Brightness of light meter. Repeat 2. (lux)	Average brightness of light. (lux)
10	10o - Repeat 1: 968, Repeat 2: 972, Average: 970 lux (results depend on the lamp and polarizers used)	2: 972, Average: 970 lux (results depend on the lamp and polarizers used)	
20	20o - Repeat 1: 882, Repeat 2: 884, Average: 883 lux (results depend on the lamp and polarizers used)	2: 884, Average: 883 lux (results depend on the lamp and polarizers used)	
30	30o - Repeat 1: 749, Repeat 2: 751, Average: 750 lux (results depend on the lamp and polarizers used)	2: 751, Average: 750 lux (results depend on the lamp and polarizers used)	
40	40o - Repeat 1: 586, Repeat 2: 588, Average: 587 lux (results depend on the lamp and polarizers used)	2: 588, Average: 587 lux (results depend on the lamp and polarizers used)	
50	50o - Repeat 1: 412, Repeat 2: 414, Average: 413 lux (results depend on the lamp and polarizers used)	2: 414, Average: 413 lux (results depend on the lamp and polarizers used)	
60	60o - Repeat 1: 249, Repeat 2: 251, Average: 250 lux (results depend on the lamp and polarizers used)	2: 251, Average: 250 lux (results depend on the lamp and polarizers used)	
70	70o - Repeat 1: 116, Repeat 2: 118, Average: 117 lux (results depend on the lamp and polarizers used)	2: 118, Average: 117 lux (results depend on the lamp and polarizers used)	
80	80o - Repeat 1: 29, Repeat 2: 31, Average: 30 lux (results depend on the lamp and polarizers used)	2: 31, Average: 30 lux (results depend on the lamp and polarizers used)	
90	90o - Repeat 1: 0, Repeat 2: 0, Average: 0 lux (results depend on the lamp and polarizers used)	2: 0, Average: 0 lux (results depend on the lamp and polarizers used)	

Graph



· Graph: plot average brightness (lux) on the y-axis against the angle between the polarizers (0-90°) on the x-axis. The curve falls from a maximum when the polarizers are parallel (0°) to zero when they are crossed (90°).

Conclusion

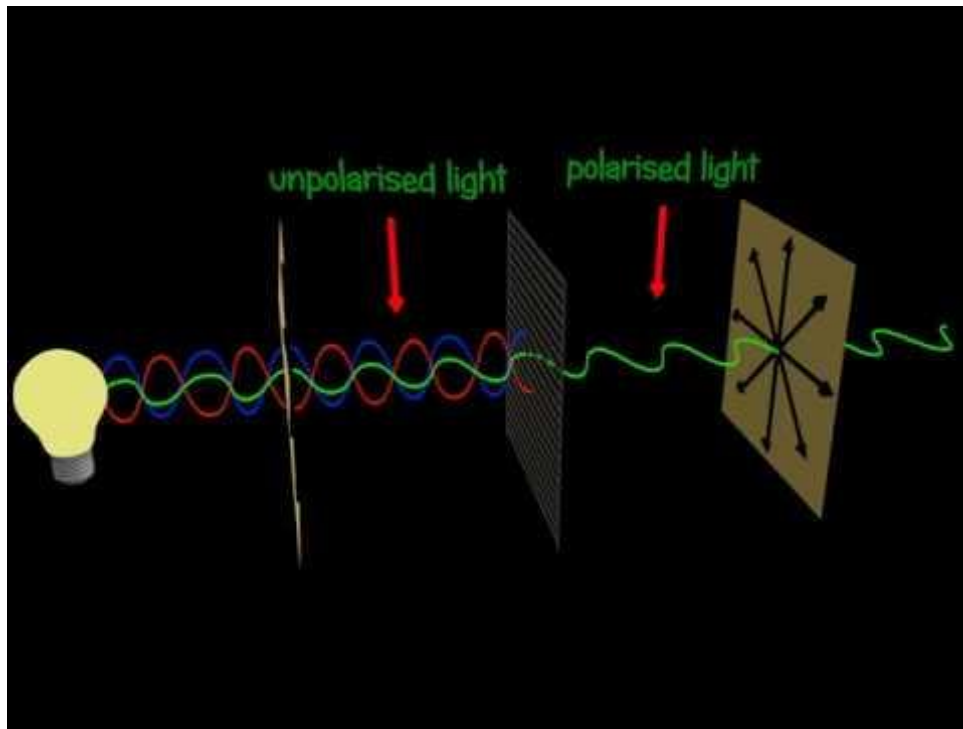
The greater the angle between the polarizers the **greater / lower** the intensity of light that passes through.

Two parallel polarizers **transmit / absorb** light

Two 'crossed' polarizers (perpendicular) **transmit / absorb** light.

Extension

How do polarizers work? Here is a video to explain how.

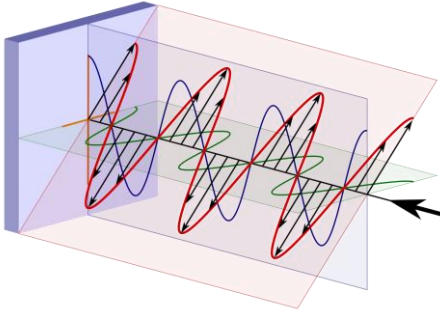


(<https://www.youtube.com/watch?v=11w32FwMvbl>)

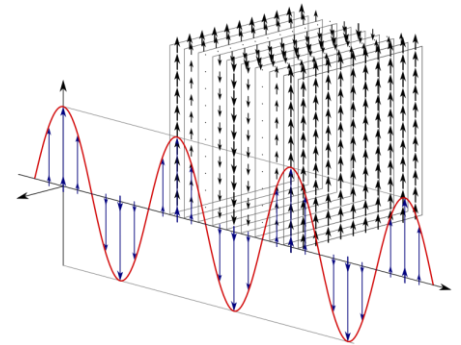
Non-polarized Light, such as sunlight vibrates in all possible directions, but polarized light vibrates in one plane only.

9P 5c - More Polarization

1. When a light wave vibrates in all directions, the light is un - polarized .



2. When a light wave vibrates in a single plane, it is polarized .



3. A Polaroid or Polarizing filter polarizes light by only allowing light in one planeto pass and absorbing light vibrating in all other directions.

A polarizing filter will: (delete the incorrect answer)

- a. twist all the wave vibrations so that they line up in a single plane
- b. block part of the vibrations while letting through only those in a specific plane

4. Filters allow light to pass through. Polaroid filters are very selective about the orientation of the light vibrations that are allowed through. The light that passes through a Polaroid filter is vibrating in a direction that is parallel to the orientation of the polarizing filter.

5. Will light pass through this pair of polarizing filters?

No. - the two filters are crossed (their axes are at right angles), so all the vibrations allowed through the first filter are blocked by the second.....

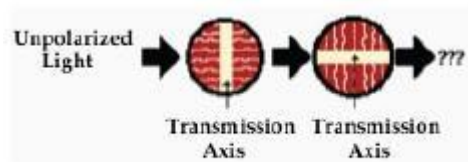


6. Will light pass through this pair of polarizing filters?

Yes - both filters have their axes in the same direction (parallel).

so the vibrations that pass through the first filter also pass

through the second



7. Passing light through a Polaroid filter is not the only way that unpolarized light can be polarized. Light is also polarized when it reflects off non-metallic surfaces. When light reflects off nonmetallic surfaces such as glass, water, or a road surface, the light is partly polarized. The reflected light consists of waves that are vibrating mostly Parallel to the reflecting surface.

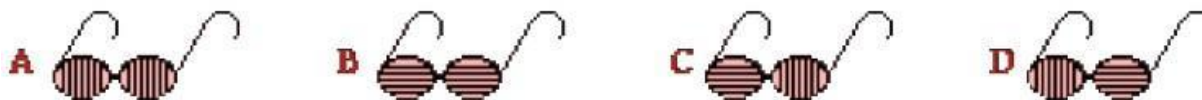
a. Parallel

b. Perpendicular

Perpendicular - the reflected glare is polarized perpendicular to the reflecting surface (horizontal glare off a flat road), so a vertically oriented polarizer blocks it.

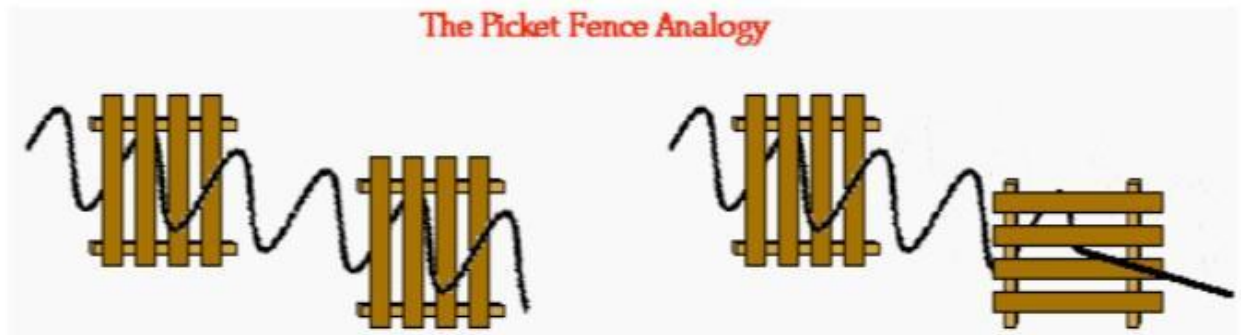
8. Mr Winde is driving down the road on a sunny day. Reflection of light off the road surface results in a large amount of glare. The glare is polarized in the horizontal plane. Annoyed by the glare, Mr Windle pulls out his Polaroid sunglasses. How must the axes of polarization be oriented in order to block the glare? (Note: the lines on the filters below represent the axis of polarization.)

The glare is polarized in the horizontal plane (parallel to the road surface). Mr Winde's polarizing sunglasses have vertical transmission axes, so they absorb the glare and he can see clearly.



Answer: ...The axes of polarization of the sunglasses must be vertical (perpendicular to the horizontal glare), so the horizontal glare vibrations are blocked.

9. The picket fence analogy is often used to explain Polarization. Use the picket fence analogy to explain your answers to questions #5 and #6. Make reference to the diagrams below in your explanations.



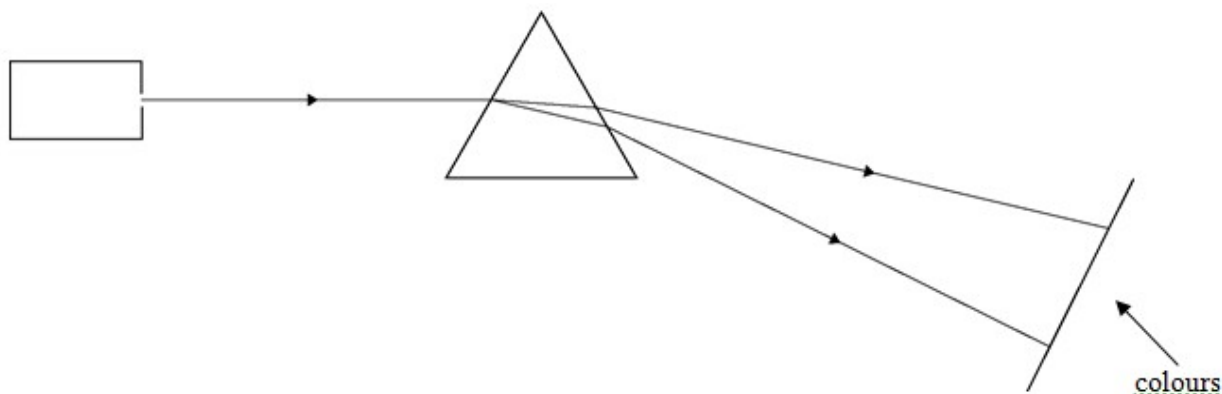
Q5. (left hand diagram)	Q6. (right hand diagram)
Q5: The two fences are crossed at 90 degrees, so the wave that squeezed through the first fence cannot pass the second. - just like a rope wave cannot pass pickets. Light waves vibrate up and down in directions at right angles to the direction of travel. A polarizer acts like a picket fence: only vibrations that are parallel to the pickets themselves hit the fence and pass through. When two polarizers are crossed at 90 degrees the second polarizer blocks all light that passes through the first. no light passes and the view is black.....	Q6: Both fences have their pickets in the same direction, so the wave that passed the first fence lines up with the gaps in the second fence and passes through. Light waves vibrate in all directions perpendicular to the direction of travel. Light that is not polarized (like light from the sun) is a mixture of all these vibrations. A polarizer acts like a picket fence: only vibrations that are parallel to the pickets themselves hit the fence and pass through. When two polarizers are crossed at 90 degrees the second polarizer blocks all light that passes through the first. no light passes and the view is black.....

When two polarizers are crossed at 90 degrees the pickets of the second fence block all remaining vibrations, so no light gets through.

9P 6a Splitting light using Refraction

How does a prism split light into different colours?

Each colour of light travels at a slightly different speed in the glass, so each colour is refracted (bent) by a different amount and the colours spread out into a spectrum.



Label the **ray box**, **prism** and **screen** on the diagram above..

· Diagram labels: ray box (light source) on the left, triangular prism in the middle, screen/paper on the right where the spectrum is produced.

Method:

- 1 Set up the apparatus as shown in the diagram above.
- 2 Shine a ray of light through the prism until you see a spectrum of colours produced on the screen or the paper.
- 3 Label on the diagram above, what colours you can see and then fill in the table of results.

· Colours labelled on the screen in order from least to most refracted: red, orange, yellow, green, blue, indigo, violet.

Results:

· Results table: the colours seen on the screen, in order - red, orange, yellow, green, blue, indigo, violet.

Colour	Results table (Colour column): red, orange, yellow, green, blue, indigo, violet - in order of least to most refracted.

Refracted Least

↓

Refracted Most

Conclusion:

- 1 Which colour was refracted the least? Red was refracted the least.
- 2 Which colour was refracted the most? Violet was refracted the most.
- 3 What are the seven colours that white light can be split into?
Red, orange, yellow, green, blue, indigo and violet.
- 4 If you mix these 7 colours of light, what colour does it give? White light.

Extension Activities:

1. Research and find out why refraction is able to split white light into seven colours.
2. Which of the colours has the longest wavelength? Red has the longest wavelength.
3. Which of the colours has the shortest wavelength? Violet has the shortest wavelength.
4. How is a rainbow formed and why is it that shape?
5. Explain with a diagram, how a rainbow forms.

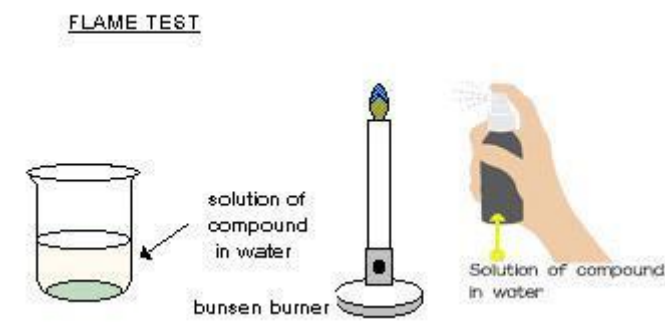
White light is a mixture of colours and each colour has a different wavelength, so glass refracts each one by a different amount - spreading them into the spectrum.

Sunlight is refracted and internally reflected inside raindrops, splitting it into colours; the drops sending light to your eye lie on a cone at about 42 degrees, so the rainbow is an arc.

· Diagram: sunlight enters a raindrop, refracts (dispenses into colours), reflects off the inside back of the drop, then refracts again as it leaves - red bends least, violet most.

9P 6b Flame Testing and Emission Spectra

Flame Testing Practical



1. For each compound, identify the metal element present.
2. Lightly spray each solution into the roaring blue bunsen burner flame and observe the colour of the flame.

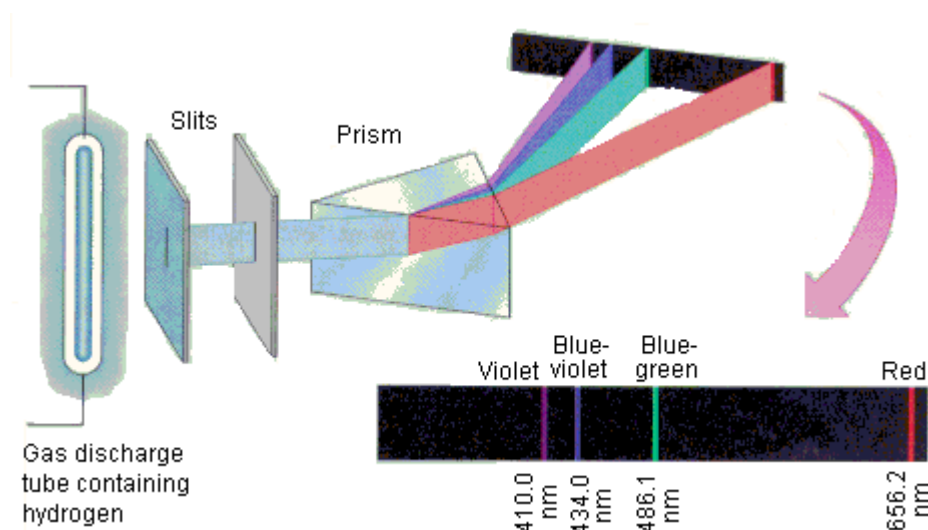
Identify the metal from the flame colour: sodium - yellow/orange, potassium - lilac, lithium - crimson, calcium - orange-red, copper - blue-green, barium - apple green.

Compound	Metal Element	Colour
		Flame colours: lithium - crimson red; sodium - yellow/orange; potassium - lilac; calcium - orange-red (brick red); copper - blue-green; barium - pale green.
X	Compound X is identified from its flame colour: yellow = sodium, crimson red = lithium, lilac = potassium, blue-green = copper. Record the colour and match it to the element.	

For Mystery compound X, conduct a flame test and identify the metal element contained in it.

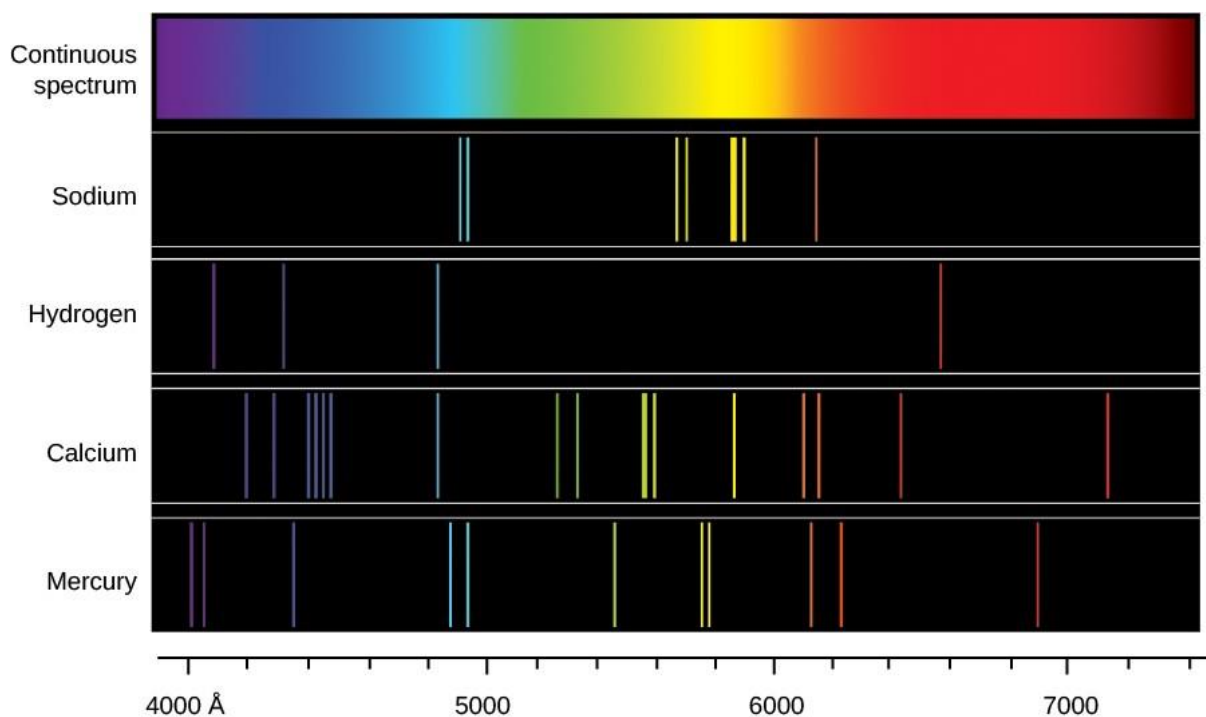
Emission Spectra

- The colours you observed above are actually a combination of a specific set of colours mixed together.
- The set of colours that are produced can be separated (using a prism or diffraction grating) to form a line emission spectra.
- The line emission spectra is unique to that particular element. It can therefore be used to identify elements.



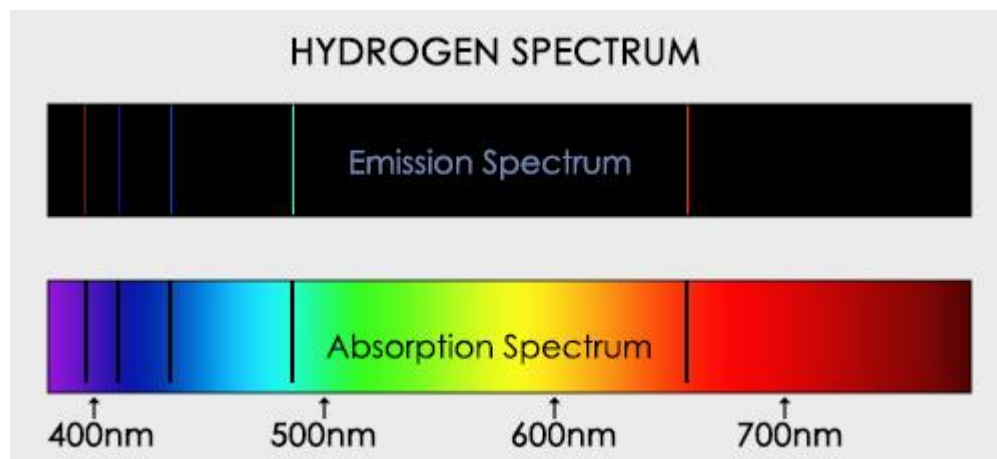
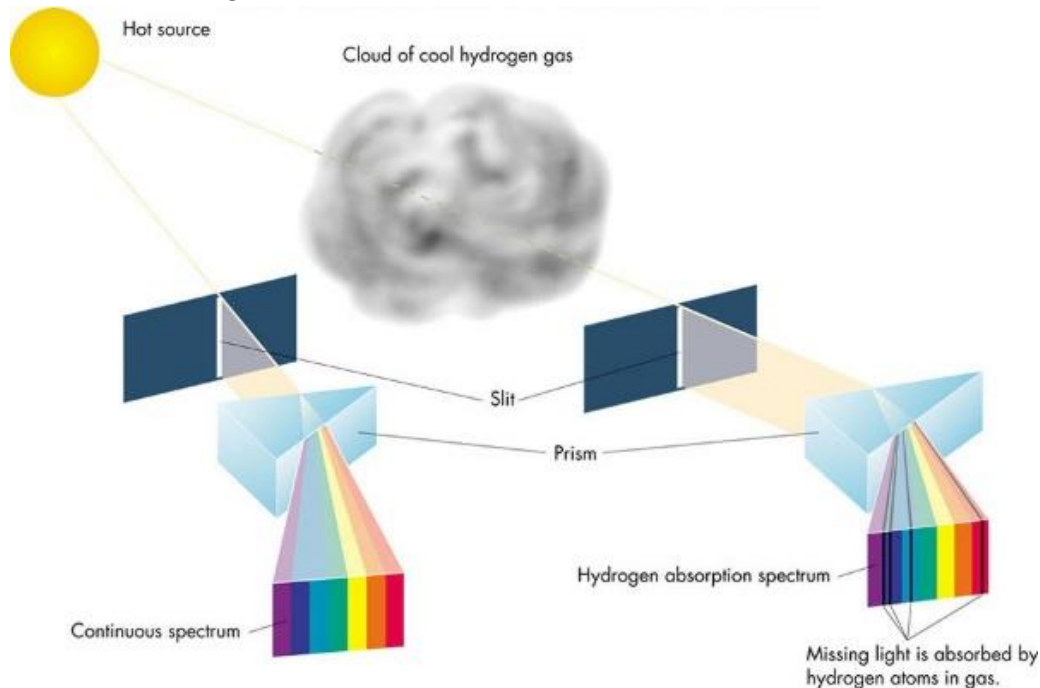
Your teacher may show you a demonstration using gas discharge tubes to form spectra for different elements.

· Gas discharge tubes: each gas (hydrogen, helium, neon, mercury, etc.) gives its own unique set of coloured lines - a line emission spectrum used to identify that element.



9P 6c Stellar Spectra

The interior of a **star** produces a continuous spectrum of light, like a rainbow. Cooler gases in the outer layers of the star absorb certain wavelengths of light, causing dark lines to appear in the spectrum. The resulting **absorption spectrum** can tell astronomers a great deal about the star.



- The lines of an emission spectra for a particular element will match up with the missing lines that are observed in an absorption spectra.
- The emission lines and absorptions lines are unique to that particular element.
- These spectra allow you to identify the element(s) present in a star. The spectra are a bit like the element's unique 'fingerprint'.

How are stars classified?

Introduction: Late in the 19th century, Harvard astronomer Edward Pickering wanted to sort and catalog the thousands of star spectra that had been collected by the Harvard Observatory. He hired several women to do the work, paying them 25 cents a day. The most prominent of these women was Annie Jump Cannon, who devised a classification system still used today.

Stars are classified by their surface temperature (colour) using their absorption spectra, into classes O, B, A, F, G, K, M - from hottest blue to coolest red.

Harvard Classification Scheme			
Class	Color	Prominent spectral lines	Surface Temp. (K)
O	Blue	Ionized helium, hydrogen	> 25,000
B	Blue-white	Neutral helium, hydrogen	11,000 – 25,000
A	White	Hydrogen, ionized sodium, ionized calcium	7,500 – 11,000
F	White	Hydrogen, ionized sodium and calcium; neutral sodium and calcium	6,000 – 7,500
G	Yellow	Neutral sodium and calcium, ionized calcium, ionized iron, ionized magnesium	5,000 – 6,000
K	Orange	Neutral calcium, neutral iron, neutral magnesium	3,500 – 5,000
M	Red	Neutral iron, neutral magnesium, and neutral titanium oxide (not shown)	< 3,500

Classification Activity

Use the Gizmo to find the elements that are present in the spectra of stars 1 through 4. Remember to check both the **Neutral spectra** and the **Ionic spectra**. Then use the table above to classify each star and describe its surface temperature. (Note: A star spectrum may not display lines of all of the elements typical of its spectral class.)

Star	Color	Elements in spectrum	Class	Surface Temperature (K)
1	Star 1: blue; elements helium and hydrogen; Class B; surface temperature 11,000-25,000 K.			
2	Star 2: white; elements hydrogen, ionized sodium and calcium; Class A; surface temperature 7,500-11,000 K.			
3	Star 3: yellow; elements neutral sodium and calcium, ionized iron and magnesium; Class G; surface temperature 5,000-6,000 K.			
4	Star 4: red; elements neutral calcium, neutral iron, neutral magnesium and titanium oxide; Class M; surface temperature below 3,500 K.			

Extension: What else can we learn from stellar spectra?

1. Observe: Observe the spectra of stars 5 – 10. Identify the elements in each spectrum, and try to classify each star. If you notice unusual features in these spectra, describe them.

Star	Color	Elements in spectrum	Class	Unusual features
5	Orange-red		K	
6	Blue		B	
7	Yellow		G	
8	White		A	
9	Yellow		F	
10	Red		M	

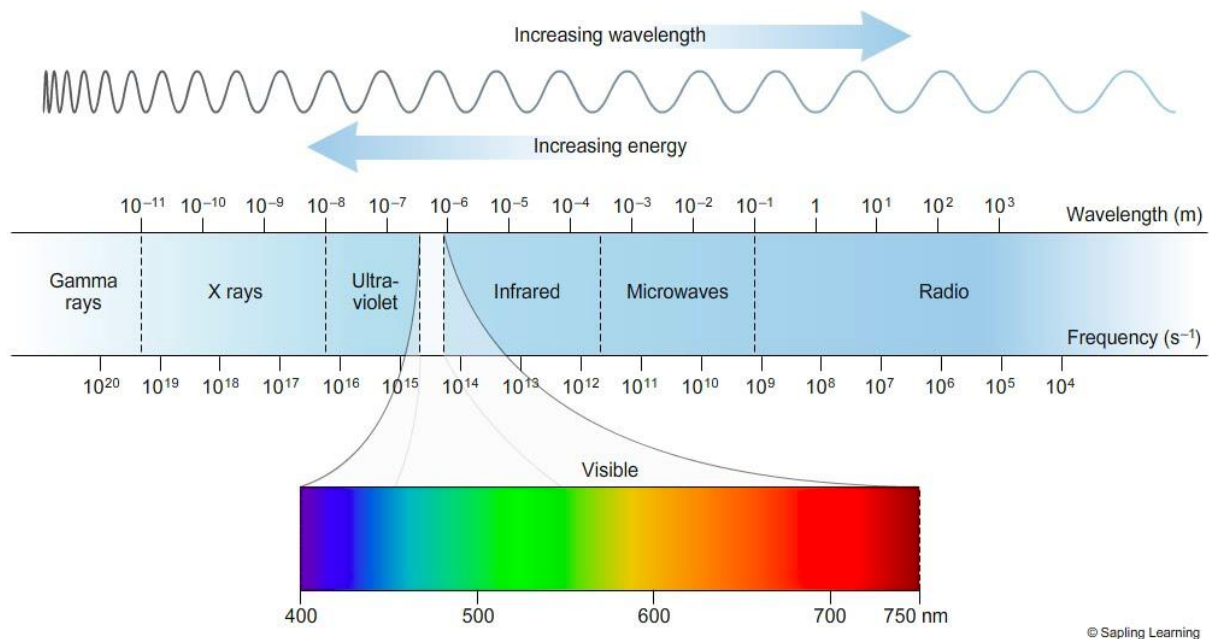
Match: Write the number of the star or object that matches each description. Then use this information to help you identify the elements and reclassify the stars in the table above.

- 5 High atmospheric pressures in a star cause spectral lines to be broadened, or “smeared out.” **Giant stars**, which have relatively low atmospheric pressures, are characterized by narrow spectral lines.
- 6 If a star is moving away from an observer, spectral lines are **redshifted**, or shifted toward the red end of the spectrum. An approaching star is **blueshifted**.
- 7 A star orbited by a large planet will move in a small circle. This will cause its spectrum to be slightly redshifted part of the time and blueshifted at other times.
- 8 **Binary stars** are pairs of stars that orbit one another. Their presence is indicated by two spectra that shift in opposite directions.
- Cepheid variable** stars change their brightness in a regular cycle. Gas pressure builds up, causing the stars to expand quickly. When the 9 pressure is released, the star contracts, and the intensity of some spectral lines may decrease.
- 10 A **nebula** is an enormous cloud of gas and dust in which stars are born. Most nebulae produce an **emission spectrum**, which is characterized by bright lines of color against a dark background. The bright lines in an emission spectrum correspond to the dark lines in an absorption spectrum.

9P 6d EM Spectrum

EM Spectrum

Visible light is just a tiny part of a much bigger spectrum called the Electromagnetic or EM spectrum. Just like visible light, the EM spectrum is made of waves of different wavelengths and frequencies.



· EM spectrum labelled left to right (increasing frequency and energy): radio, microwave, infrared, visible light, ultraviolet, X-rays, gamma rays.

- The gamma ray end of the spectrum is the high frequency, low wavelength, high energy and high danger end of the spectrum.
- The radio wave end of the spectrum is the low frequency, high wavelength, low energy and least dangerous end of the spectrum.
- All parts of the EM Spectrum travel at the speed of light which is: $3 \times 10^8 \text{ m/s}$

Extension / Homework Activity:

Pick a part of the EM Spectrum and research an application/use of this part.

e.g. Infrared is used in TV remote controls and thermal imaging cameras, because warm objects emit infrared radiation.

9P 7a - Gravity and the Solar System

Deep Learning Outcomes

1. Gravity keeps the planets in orbit around the sun.
 - a. The planets of the Solar System
 - b. Characteristics of the planets
2. How does Orbital Speed change with distance from the Sun.
3. How does Orbital Period (year) change with distance from the sun.

Part 1 - Gravity

Gravity is one of the fundamental forces in the universe. We are inside the Earth's gravitational field.

What is a Field?



A field is a region (or space) in which an object experiences a **non-contact force**.....

A Gravitational Field is a region in which a **mass**..... experiences a **non-contact force**.....

Unit 7 - The Solar System

The Solar System

SOLAR SYSTEM VIDEO - 12mins

<https://www.youtube.com/watch?v=Qd6nLM2QIWw>

Research the 8 planets of the Solar System.

The **distance from the sun to earth** is called an **Astronomical Unit (AU)** and is a useful unit in astrophysics.

Name of planet	Rocky or Gas?	Distance from the Sun (AU)	Speed (km/s)	Year (days or years)	Interesting fact(s)
Mercury				88 days	Venus: Rocky; 0.72 AU; 35.0 km/s; 225 days; hottest planet (about 465 C) because of its thick atmosphere.
Earth		1.0			Mars: Rocky; 1.52 AU; 24.1 km/s; 687 days; the red planet (iron oxide dust); has the biggest volcano, Olympus Mons, and 2 moons.
(Pluto) (no longer a planet)	Rocky	39.5	4.7	248 years	It's moon is called Charon (the ferryman of Hades)

· Pluto (dwarf planet): Rocky; 39.5 AU; 4.7 km/s; 248 years; its moon is Charon.

· Jupiter: Gas; 5.2 AU; 13.1 km/s; 11.9 years; largest planet, with the Great Red Spot storm and about 95 moons.

· Saturn: Gas; 9.5 AU; 9.7 km/s; 29.5 years; famous for its bright ring system and has the most moons (about 146).

· Uranus: Gas; 19.2 AU; 6.8 km/s; 84 years; rotates on its side (98 degree tilt), so it rolls around the Sun.

· Neptune: Gas; 30.1 AU; 5.4 km/s; 165 years; the windiest planet, with winds up to about 2,100 km/h.

Unit 7 - The Solar System

Model Solar System

Can you make a scale model of the solar system? Measure the masses of pieces of plasticine to the correct proportion, taking the Earth to be 1g. Do **only one** of each planet for the whole class.

<u>Planet</u>	<u>Mass of planets relative to the mass of the Earth</u>	<u>Mass of plasticine / grams</u>
Mercury	0.05	0.05
Venus	0.8	0.8
Earth	1	1
Mars	0.1	0.1
Jupiter	300	300
Saturn	100	100
Uranus	15	15
Neptune	17	17

Extension

[Scale Calculator](#)

If the model earth was placed 1m away from the sun how far from the sun would the other planets be?

<u>Planet</u>	<u>Distance from the Sun (AU)</u>	<u>Distance from the Sun in your model / m</u>
Mercury	0.4	0.4
Venus	0.7	0.7
Earth	1	1
Mars	1.5	1.5
Jupiter	5	5
Saturn	10	10
Uranus	19	19
Neptune	30	30

Can you put your planets in the correct relative spacing?

- You might just fit this into the corridor.
- Put each planet on a piece of paper so that you do not stand on them.

Yes - place each planet at a scaled distance from the Sun so the relative spacing is correct: Mercury closest, Neptune about 30 times further out than Earth.

Unit 7 - The Solar System

Extension - Planet Fall

1. Why was Pluto downgraded to a Dwarf Planet in 2006?



Pluto is much smaller than the planets and has not cleared its orbit - it shares the Kuiper Belt with many similar icy objects, so it was reclassified as a dwarf planet.

2. Is there anything else unusual about Pluto's orbit?

Its orbit is highly elliptical and tilted about 17 degrees to the other planets, and it crosses inside Neptune's orbit, so sometimes Pluto is closer to the Sun than Neptune.

3. How can you remember the order of the planets:

Use a mnemonic sentence, e.g. 'My Very Easy Method Just Speeds Up Naming' = Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.

4. Why is Venus the hottest planet in the Solar System when Mercury is closer to the sun?

Venus has a thick carbon dioxide atmosphere that traps heat (runaway greenhouse effect), so at about 465 C it is hotter than Mercury, which has almost no atmosphere.

Unit 7 - The Solar System

Part 2 - Orbital Speed

What happens to the speeds of the 8 planets as you go further from the sun?

Plot a graph to see the relationship. Put Distance from the sun on the x-axis and orbital speed on the y-axis.

Try using Google Sheets to produce your graph.

There is a video on ISLE of a demo of using Google Sheets.

Results Table

The orbital speed gets smaller (decreases) as the planets get further from the Sun - the closer a planet is, the faster it moves.

Name of planet	Distance from the Sun (AU)	Speed (km/s)
Mercury		
Earth	1.0	

Results table (distance AU / speed km/s). Mercury 0.39 / 47.4; Venus 0.72 / 35.0; Earth 1.0 / 29.8; Mars 1.52 / 24.1; Jupiter 5.2 / 13.1; Saturn 9.5 / 9.7; Uranus 19.2 / 6.8; Neptune 30.1 / 5.4.

Graph

Conclusion

Describe the trend. Use ideas like Gravitational Energy and Kinetic Energy to explain your conclusion.

The further a planet is from the Sun, the slower its orbital speed (the graph curves downwards). Distant planets have gained lots of gravitational potential energy and have little kinetic energy, so they move slower.

Unit 7 - The Solar System

Part 3 - Orbital Period

Orbital period is the time it takes for the planet to orbit the sun once. We also call this the year. For Earth, the year is 365 days.

How does the orbital period of the 8 planets change with the distance from the sun?

Plot a graph of Orbital Period (in earth years) on the y-axis against distance from the Sun (in AUs) on the x-axis.

Try using Google Sheets.

Results Table

The orbital period increases with distance from the Sun - the further the planet, the longer its year (Kepler's third law).

Name of planet	Distance from the Sun (AU)	Year ('earth years')
Mercury		$(88/365) = 0.24$
		$(225/365) = 0.62$
Earth	1.0	1.0

· Results table (distance AU / year in Earth years). Mercury 0.39 / 0.24; Venus 0.72 / 0.62; Earth 1.0 / 1.0; Mars 1.52 / 1.88; Jupiter 5.2 / 11.9; Saturn 9.5 / 29.5; Uranus 19.2 / 84; Neptune 30.1 / 165.

Graph

Conclusion

Describe the trend and try to explain it. Use ideas like circumference and speed to explain your conclusion.

The further a planet is from the Sun, the longer its orbital period. A distant planet has a much bigger orbit to travel (a longer circumference) and moves more slowly, so its year is much longer.

Unit 7 - The Solar System

Questions

[Gravity and the Solar System | Science Quiz](https://quizizz.com/admin/quiz/5d631e5d655a24001e90bc28/gravity-and-the-solar-system)

<https://quizizz.com/admin/quiz/5d631e5d655a24001e90bc28/gravity-and-the-solar-system>

[Kahoot me up](https://create.kahoot.it/share/the-solar-system/2a07be85-8669-4fe6-8ade-7b34598b3c5d)

<https://create.kahoot.it/share/the-solar-system/2a07be85-8669-4fe6-8ade-7b34598b3c5d>

Extension Questions

1. Why does the Earth have seasons?
Earth's axis is tilted at 23.5 degrees, so as it orbits the Sun each hemisphere tilts towards the Sun for part of the year (summer: higher Sun, longer days) and away for the rest (winter).

2. What is the Milky Way and what time of year is it the best time of year to see it?
The Milky Way is our own galaxy - a barred spiral of hundreds of billions of stars. It is best seen in the northern hemisphere summer (July-September), looking south.

3. How many potentially inhabitable exo-planets have been found?
Around 70 exoplanets are confirmed as potentially habitable (thousands more await confirmation); the number keeps growing as new missions report.

4. Why is the Astronomical Unit a useful unit for astrophysics?
Because distances inside the Solar System are enormous in km but tiny in light years, 1 AU (the Earth-Sun distance) gives small, easy-to-use numbers.

The Moon and its Phases

Moons



Most of the planets in our solar system have moons. The number of moons that these planets have (at the time of writing), range from 0 to 95 (with more being discovered every year!).

Try to answer these starter questions in 5 minutes...

1. Which planet has the most moons?
Saturn has the most moons (about 146 confirmed at the time of writing).
2. Which planets have no moons?
Mercury and Venus have no moons.
3. How are moons formed?
Moons form from leftover gas and dust orbiting a young planet, by capture of passing asteroids, or from the debris of a giant impact (how Earth's Moon formed).

The Earth's moon is appropriately called 'the Moon' and is in orbit at a distance of 385,000 km from the Earth..

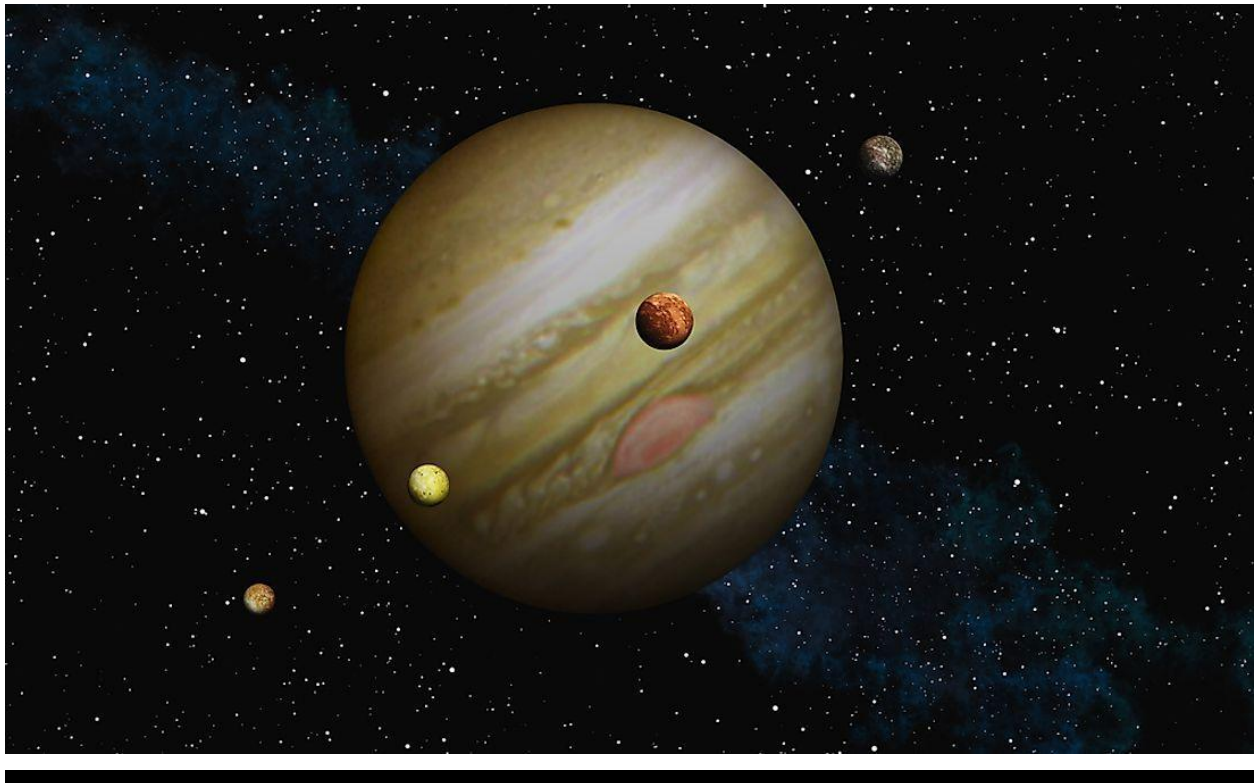
Do you know any other facts about the moon? Talk to your neighbor and discuss what each of you know about the Moon.

Unit 7.5 – The Moon

Make some notes here and report back if asked:

Moon facts: it is about 385,000 km away, has no atmosphere and its gravity is about 1/6 of Earth's. Its surface has craters, dark maria and mountains, and it is tidally locked.

Further Research of Moon Facts



Unit 7.5 – The Moon

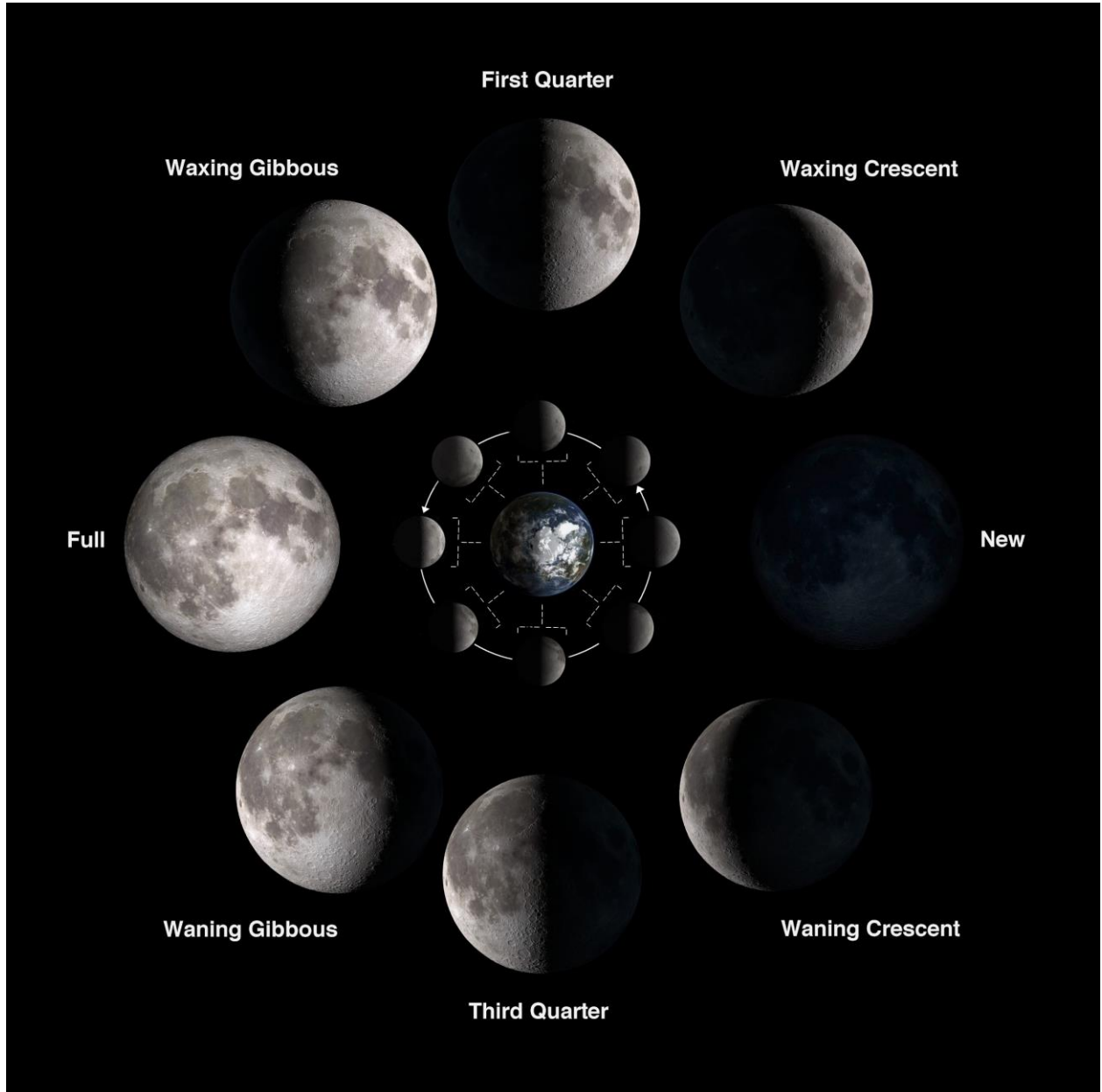
Which planet shown with some of its moons is this one?

Complete the table below (try to remember the order of the planets without looking):

Saturn - the ringed planet, shown with several of its many moons orbiting it.

Name of Planet	Number of Moons
Mercury	
· Number of moons: Mercury 0, Venus 0, Earth 1, Mars 2, Jupiter 95, Saturn 146, Uranus 28, Neptune 16 (numbers change as more are discovered).	

Phases of the Moon



The Moon looks different at different times of the month and this is because of the relative positions of the Earth, Moon and Sun. Look at this image with a neighbour and see if you can discuss and understand what it is showing.

Log on to the Gizmo to learn more about this:

<https://gizmos.explorellearning.com/>

Unit 7.5 – The Moon

Find the Gizmo entitled “Phases of the Moon” and click on “launch Gizmo”

Click on the play button and watch how the light from the sun and the position of the Moon changes the way the Moon looks from an observer on Earth.

By doing this you can understand the phases of the Moon by observing the positions of the Moon, Earth and Sun. A view of the Moon from Earth is shown on the right as the Moon orbits Earth. Learn the names of Moon phases and in what order they occur. Click Play to watch the Moon go around, or click Pause and drag the Moon yourself.

After you have had a play try to complete the Exploration sheet. Your teacher may add this to your google classroom or you may be able to access this on the Gizmo site.

Practical Work

By using a lamp (the Sun) and a small white ball (the moon). Your head can be the Earth!. See if you and a partner can simulate the motion of the moon. Try to see if you can identify the 8 phases of the moon.

Watch this video [Moon Phases Demonstration](#)

This will give you an idea of how to carry out your practical work.

If you can try to take a photo of the moon (the small ball) in its eight phases. If time you can test yourself by drawing and labelling the eight phases of the moon below-you could use your photos to copy from if they are good images:

The eight phases in order: new moon, waxing crescent, first quarter, waxing gibbous, full moon, waning gibbous, last (third) quarter, waning crescent.

The many moons in our Solar System are all very different. Try to find a moon that has some unique qualities and do a little research.

Jot down some fun facts below and be ready to share

Fun facts: the Moon is tidally locked so we always see the same face; with no atmosphere its footprints last for millions of years; 12 people have walked on it.

Unique Moon fun facts

Unique moon facts: Io is the most volcanically active body in the Solar System; Europa may hold a hidden ocean; Titan has lakes of liquid methane and a thick atmosphere.



9P 8a Galaxies and The Universe

Galaxies

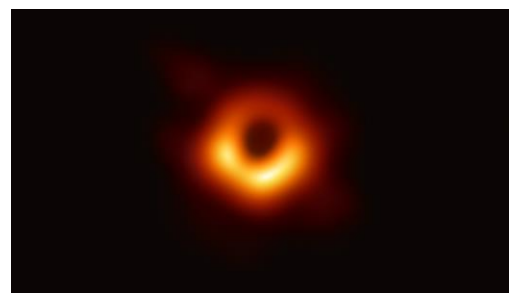


Answer the following questions in full sentences:

1. What is a galaxy?
A galaxy is a huge collection of billions of stars together with gas, dust and dark matter, held together by gravity.
2. The sun is part of which galaxy?
The Sun is part of the Milky Way galaxy.
3. What are the 3 main categories of galaxy?
Spiral, elliptical and irregular.
4. What is the Universe?
The Universe is everything that exists - all matter, energy, space and time, including every galaxy, star and planet.

FunFact:

In recent years we have discovered that there is a supermassive black hole at the centre of almost all galaxies, including our own.



9P 8b Redshift

Redshift and The expanding Universe

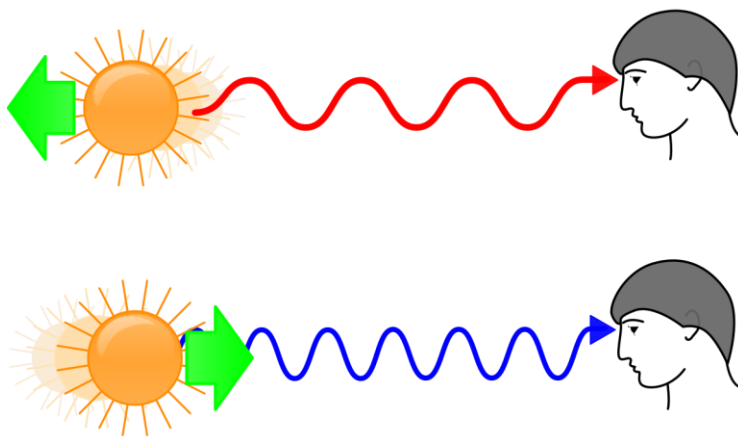
When the source is moving away from you, each wave is emitted further away than the wave before. This means that the wavelength of the wave is now

longer and since the speed is the same and speed = frequency x wavelength the observed frequency is therefore lower, and moves

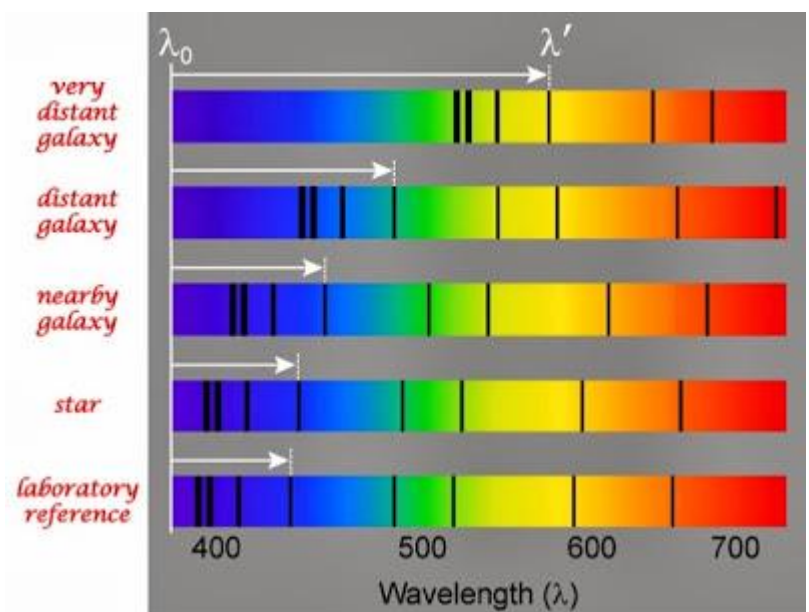
towards the red part of

the visible spectrum. This

is called redshift.



When we measure the light from other galaxies we see that the hydrogen lines have a Red-Shift!



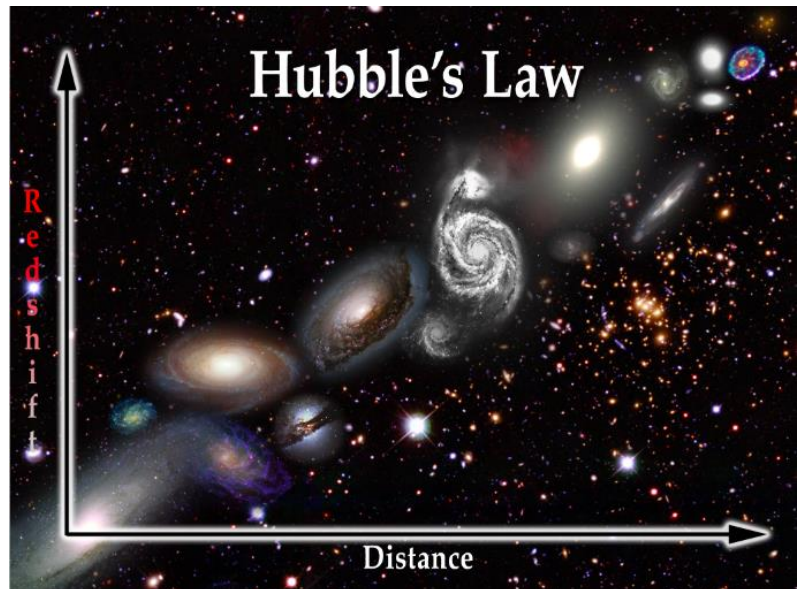
9P 8c Hubble's Law

Aim:

In this investigation you will be using real scientific data to investigate the relationship between **recessional velocity of galaxies** and the **distance** they are away from us.

Instructions:

1. Access your Google Sheet entitled 'Hubble's Law'



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https://docs.google.com/spreadsheets/d/1PqGcEXP4fpODsLUDUAZid6jllq76oIV2_YTBgcs3nFA/edit#gid=0

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Hubble's Law ☆

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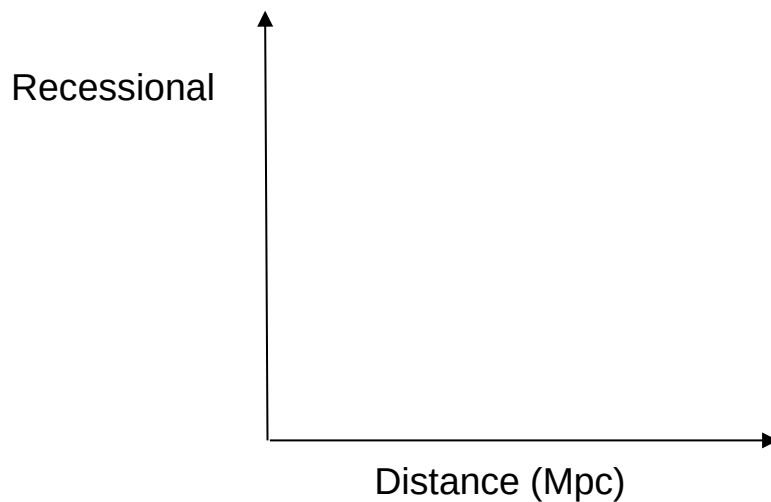
	A	B	C	D	E	F	G	H	I	J	K
	Galaxy	Calcium K	Calcium K Red Shift	Calcium H	Calcium H Redshift	Hydrogen Alpha	Hydrogen Alpha Red Shift	Average Red Shift	Recessional Velocity (km/s)	Angular Size	Distance (light years)
1											
2	NGC 1357	3962.9	0.0074230	3998.6	0.0075847	6609.8	0.0071616	0.0073898	2216.9	1.073	
3			-1.0000000		-1.0000000		-1.0000000	-1.0000000			
4			-1.0000000		-1.0000000		-1.0000000	-1.0000000			
5			-1.0000000		-1.0000000		-1.0000000	-1.0000000			
6			-1.0000000		-1.0000000		-1.0000000	-1.0000000			
7			-1.0000000		-1.0000000		-1.0000000	-1.0000000			
8			-1.0000000		-1.0000000		-1.0000000	-1.0000000			
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+ Sheet1

2. Watch the screencast showing how to collect the data and record it in your own spreadsheet.
3. Here is the link to the data: [Hubble's Law Data](#).
4. Collect data for 7 galaxies.

Analysis:

Plot a line graph of recessional velocity v distance. There is another screencast about how to produce a line graph.



Conclusion:

What is the relationship between the recessional of galaxies and how far they are away from us?

The further away a galaxy is, the faster it is receding (moving away) from us.

Recessional velocity is directly proportional to distance - this is Hubble's Law ($v = H_0 \times d$).

So the line graph is a straight line through the origin with a positive gradient.

Can you explain this relationship?

Because the Universe is expanding: space itself stretches, so the light from distant galaxies is stretched to longer wavelengths (redshifted) and they seem to recede faster.

Extension:

1. Calculate the gradient of the graph.
2. The gradient of the graph is the Hubble Constant, H_0 .
3. The value of the Hubble Constant is 67.4 km/s/Mpc. How does your value compare?

Gradient = rise / run = change in speed (km/s) divided by change in distance (Mpc), read off the line of best fit. Gradient = $(y_2 - y_1) / (x_2 - x_1)$, with units km/s/Mpc.

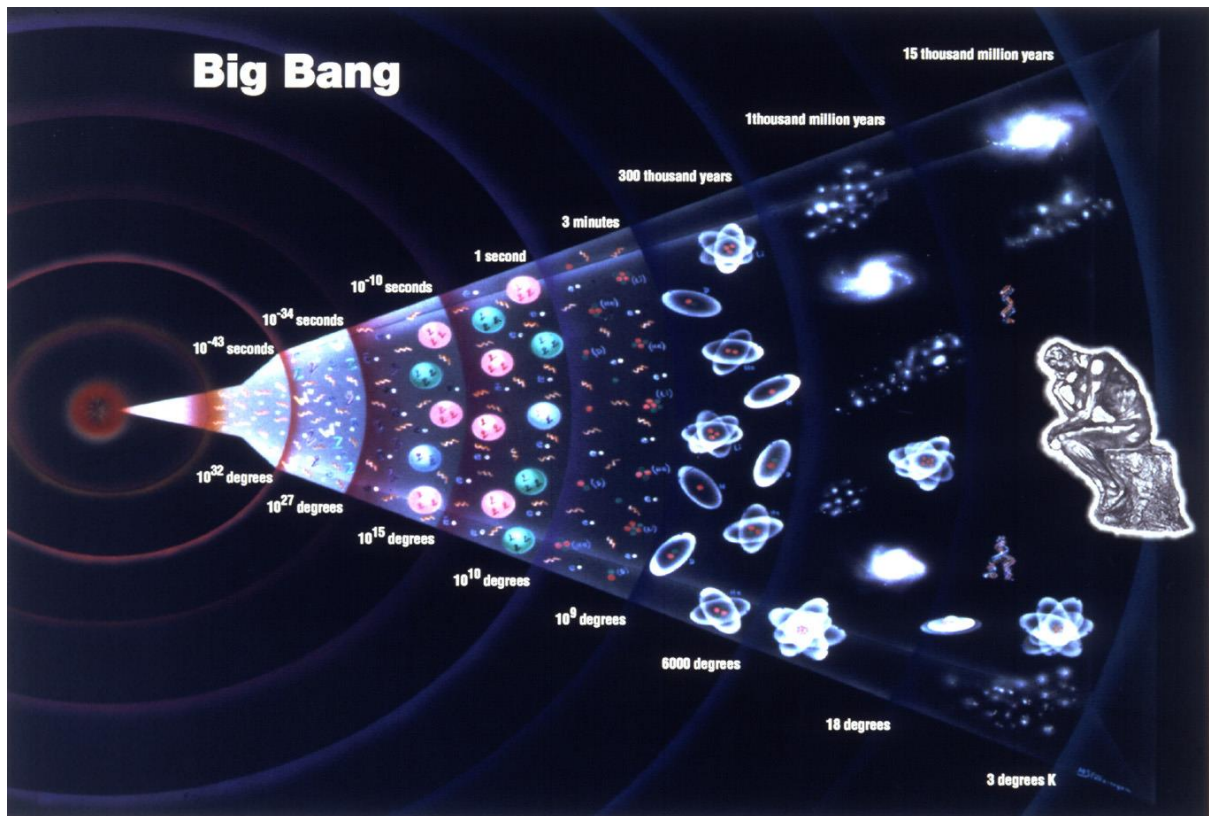
My gradient should be close to 67.4 km/s/Mpc; it is (slightly higher / slightly lower) than the accepted value, because the points do not lie exactly on a straight line.

Extended Extension:

1. Comment on the accuracy of your gradient value compared to the accepted value of the Hubble Constant. What could have caused any inaccuracy?

It is reasonably accurate if it is close to 67.4 km/s/Mpc. Inaccuracy is caused by drawing the line of best fit by eye, inaccurate readings of speed and distance, and the large uncertainties in measurements of very distant galaxies.

9P 8d The Big Bang Theory



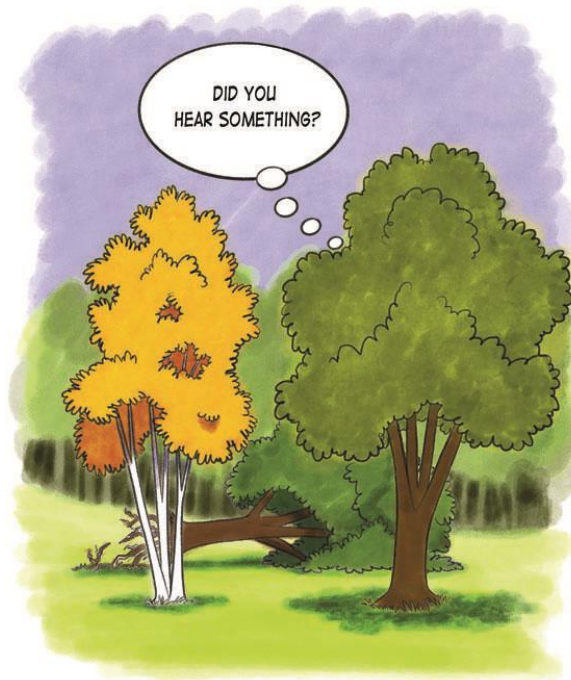
Watch the video and take some notes in the space below:

Big Bang notes: About 13.8 billion years ago all matter and energy existed in a very small, very hot, very dense point. It expanded suddenly and has been expanding ever since; the Universe is still expanding today (red shift of distant galaxies) and the cosmic microwave background is leftover radiation.

Sound and How We Hear it

One of our most useful senses is our ability to hear sound. But what is sound? A common question that confuses people is.... “When a tree falls in a forest with no one around, does it make a sound?”

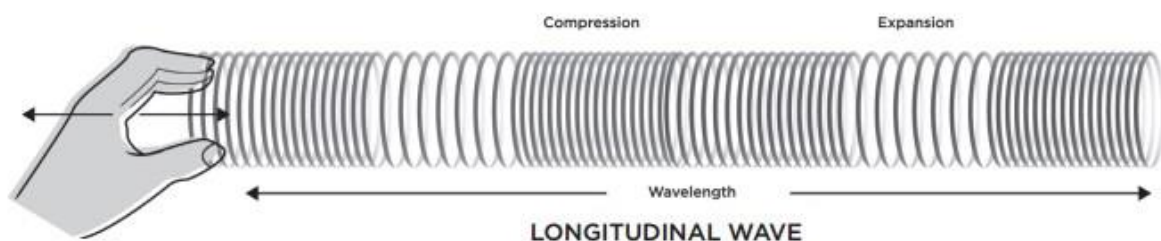
Sound is energy produced by a vibrating object. It travels as a longitudinal wave through a medium (air, water or a solid) and is detected by our ears.



Talk to your neighbour and try to come up with an explanation of what sound is, and try to discuss the tree in a forest.

Watch 'What is sound' [1. What is sound-powerpoint.ppt](#)

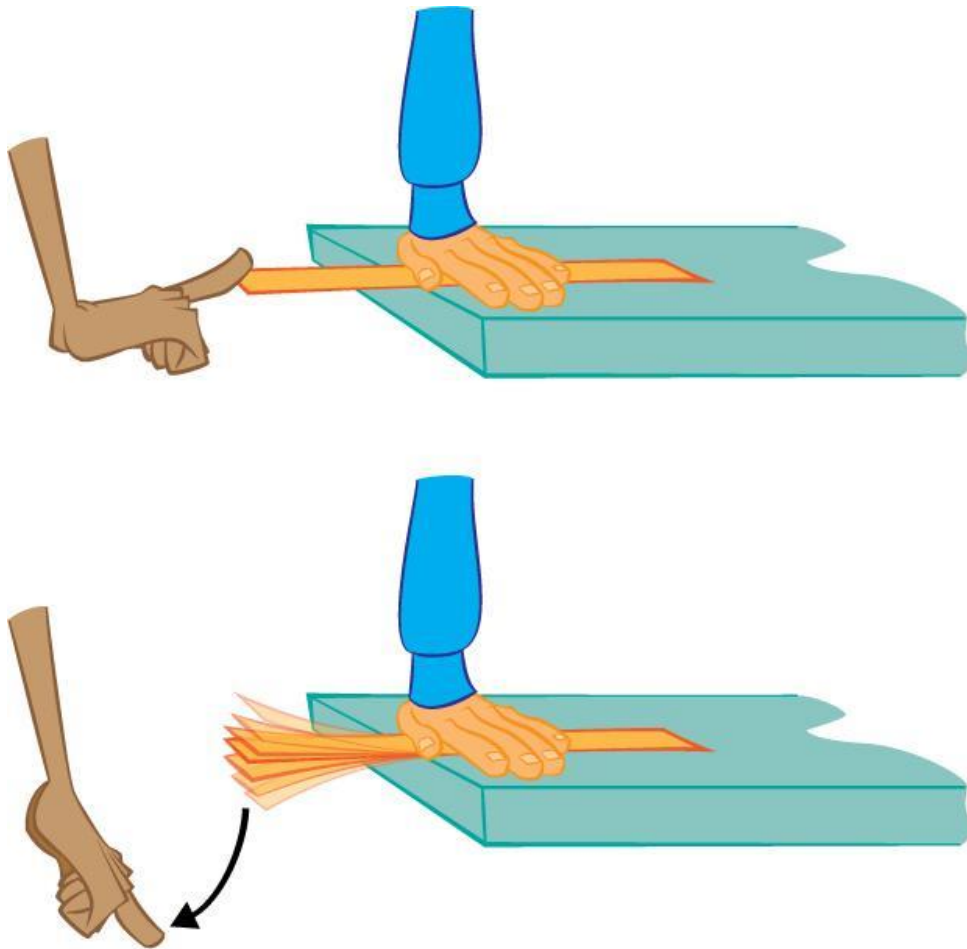
This is a Slinky and represents how sound travels through a medium.



Your teacher may let each pair take turns in producing a wave as shown above. This is called a longitude wave and is a demonstration of how air particles behave when transmitting sound.

If you place a small piece of tape in the centre of the spring you can get a better understanding of how the air particles vibrate (or oscillate)

Grab a 50cm ruler, or use your own and try to use it like below to create vibrations and make a sound. Can you make different sounds with the same ruler?



Make vibrations with a ruler, and hear how vibrations make sounds.

Yes. Changing how much of the ruler sticks over the edge of the desk changes the sound: a shorter overhang vibrates faster and gives a higher pitch, a longer overhang gives a lower pitch. Hitting it harder makes the sound louder but not higher in pitch.

Longitudinal Waves

Fill the gaps to complete the paragraphs describing the properties of longitudinal waves.

Longitudinal waves vibrate, or oscillate , in the same direction as the energy they

are transferring. They transfer energy parallel to the direction in which the waves oscillate.

Longitudinal waves can be modelled by pushing a spring backwards and forwards .

The energy is transferred horizontally along the spring, in the same direction

the waves move in. The spring returns to its original position once the energy has been

transferred because waves do not transfer matter , only energy.
waves are an Sound

example of longitudinal waves.

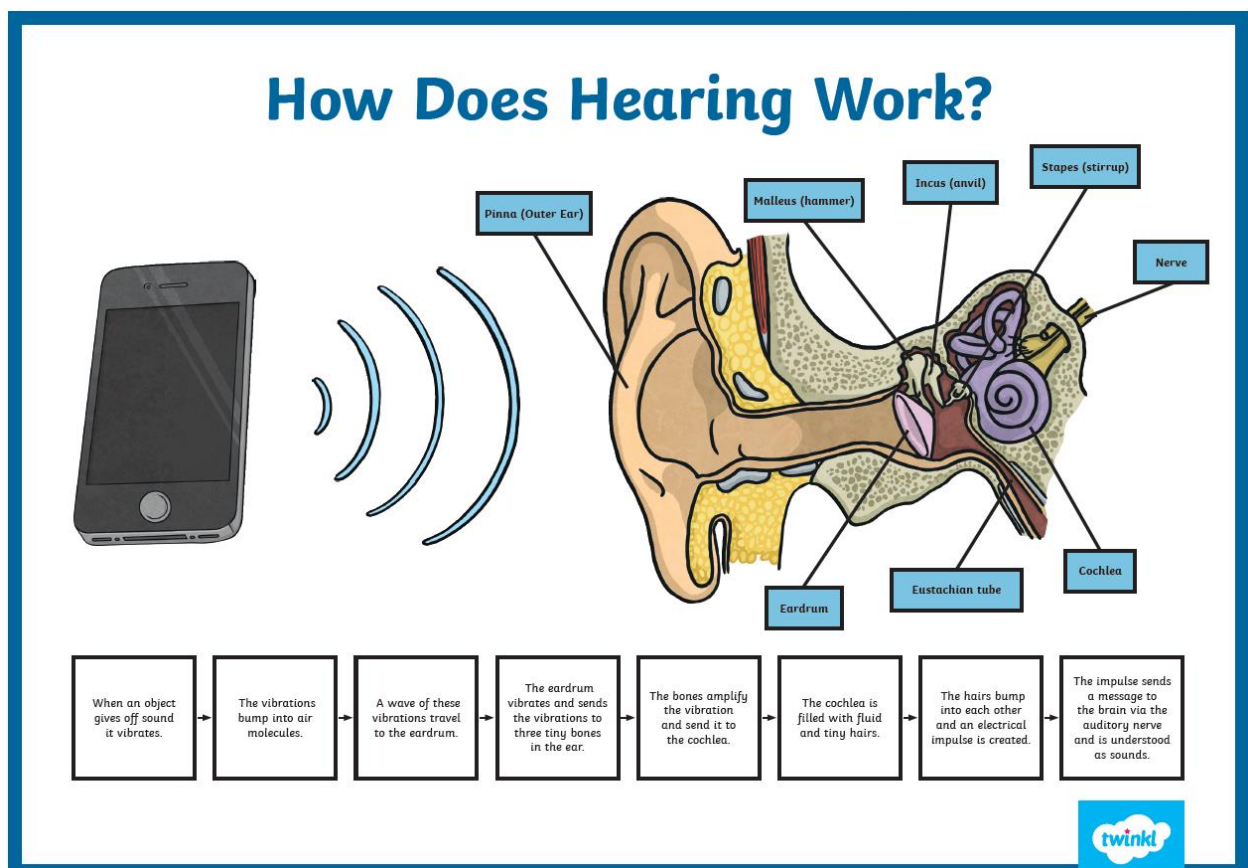
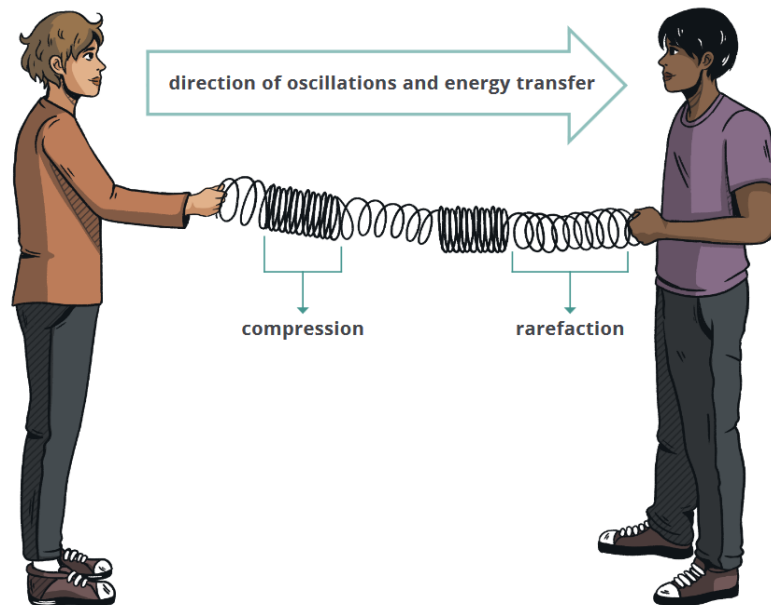
Longitudinal waves contain areas where the particles are closer together than average .

These areas are called compressions . On a spring, areas of compression are

shown where the coils are closer together. The places on the spring where the coils are

stretched out model rarefactions . These are areas on a longitudinal wave where

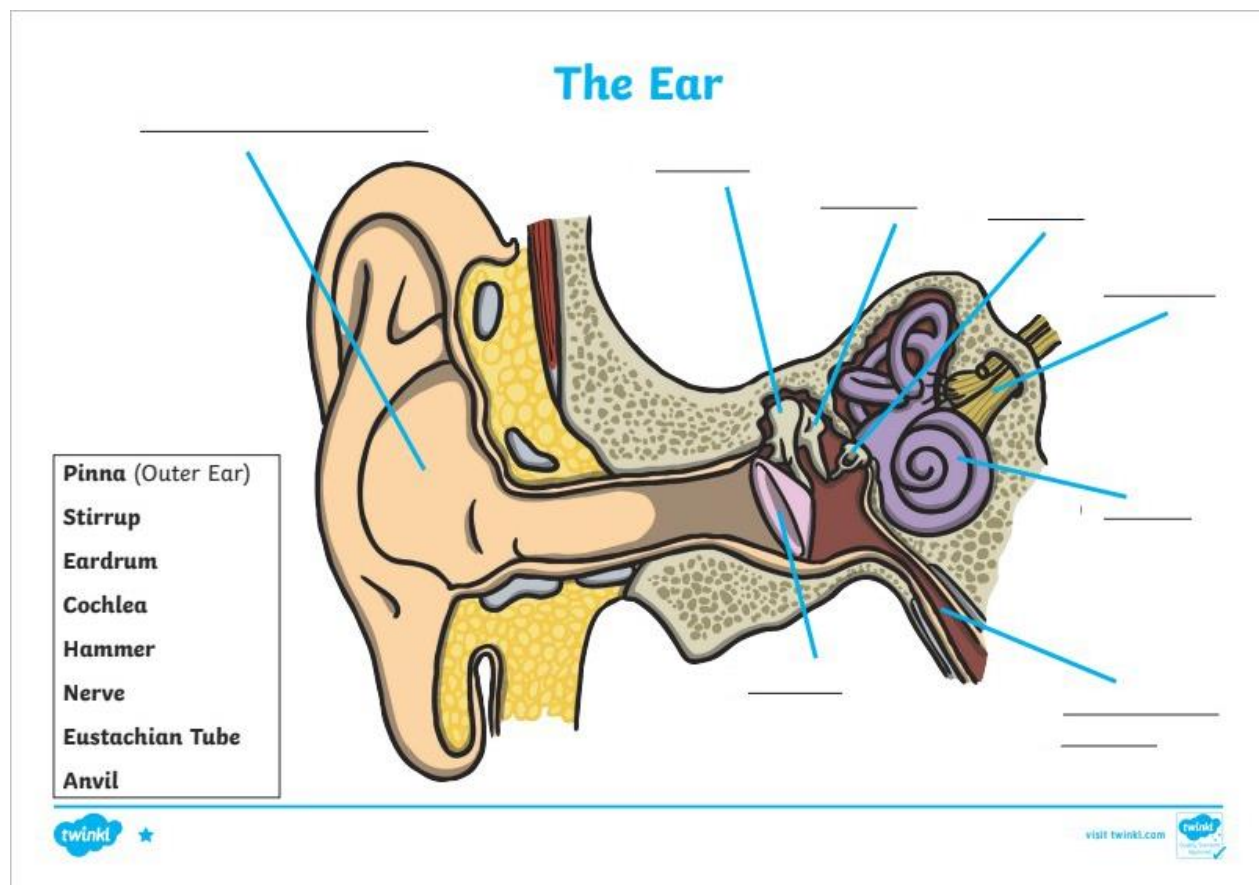
the particles are further apart than average.



After watching [4. Parts of the Ear-powerpoint.pptx](#) try to fill in the gaps below:

Fun fact: The outer ear (or the Pinna) is one of the only two parts of the human body that never stop growing. What do you think the other part is?

· Label the gaps on the ear diagram: pinna, ear canal, eardrum, ossicles (hammer, anvil, stirrup), cochlea, auditory nerve, semi-circular canals, Eustachian tube. The pinna collects sound, the canal carries it to the eardrum, the ossicles amplify the vibration, the cochlea turns it into electrical signals and the auditory nerve takes them to the brain.



Research further what each of these parts do and find out what can go wrong with the human ear. How has technology come to the rescue?

Further notes on the function of the parts of the ear:

Pinna - collects and funnels sound into the ear. Ear canal - carries the sound to the eardrum. Eardrum - vibrates when sound hits it. Ossicles - amplify and pass on the vibration. Cochlea - turns vibrations into electrical signals. Auditory nerve - carries the signals to the brain.

What problems can occur in the human ear and how can they be fixed?

Problems: ear infections / glue ear, wax blockage, burst eardrum, damaged hair cells from very loud noise, age-related hearing loss. Fixes: antibiotics and grommets, ear syringing, and technology such as hearing aids (amplify the sound) or cochlear implants (send signals straight to the auditory nerve).



[9. Whats Your Idea Hearing Aids PowerPoint.pptx](#)

<https://docs.google.com/presentation/d/1KZO7iOym5CAZ3Uuf1T4zmQ6MvGwEZXI0/edit#slide=id.p1>

Sound travels only when there is a medium to travel in. An example of a medium is air, or water, or metal etc.

Can you think of the name of something that sounds CANNOT travel through?

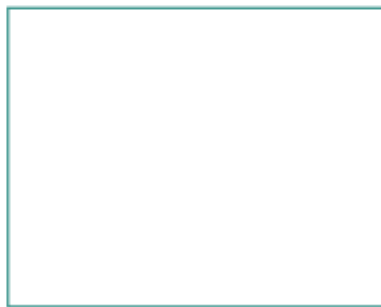
A vacuum - for example outer space. Sound cannot travel through a vacuum because there are no particles to vibrate and pass the energy on.

Wave Speed in Solids, Liquids and Gases

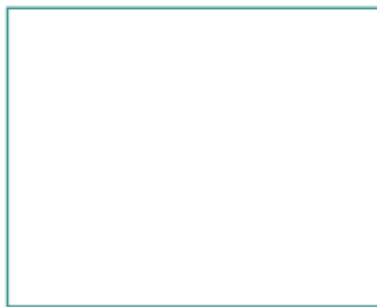
Energy is transferred as sound when particles vibrate against each other. Sound waves require particles to travel. The substance that a sound wave travels through is known as a Medium.

The three states of matter are: solid, liquid and gas. The particles in each of the states of matter are arranged differently.

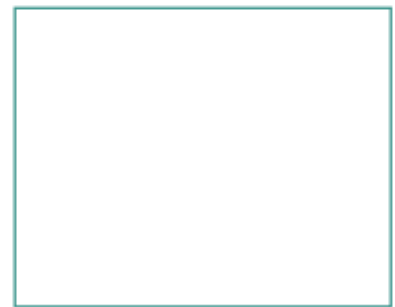
1. Draw the particle models for a solid, a liquid and a gas in the boxes below.



solid



liquid



gas

2. Compare the arrangement of the particles in solids, liquids, and gases.

Solid - the particles are very close together and tightly packed in a regular (lattice) arrangement.

Liquid - the particles are close together but arranged randomly and can slide past each other.

Gas - the particles are far apart and randomly arranged, moving freely in all directions.

So the distance between the particles increases from solid to liquid to gas, and the forces between them get weaker.

3. Which state of matter do sound waves travel through most quickly? Explain your answer.

Solids. The particles are closest together and strongly bonded, so vibrations are passed on from particle to particle quickly. In gases the particles are far apart, so energy is only transferred when they collide, which is much slower.

Unit 9.5 – 9.6 - Sound

Sound travels fastest in solids because the particles are packed tightly and strongly bonded, so vibrations pass on quickly.

It is slower in liquids, where the particles are close but not fixed in position, and slowest in gases.

The closer together the particles are, the faster sound travels through the material.

Outer space is described as a vacuum. There are no particles in a vacuum.

4. Explain why sounds cannot be heard in outer space.

Sound is a vibration and needs particles to pass the energy on from one particle to the next.

Outer space is a vacuum, so there are no particles between the stars to vibrate.

With no particles, the vibrations cannot be transferred, so no sound wave can travel.

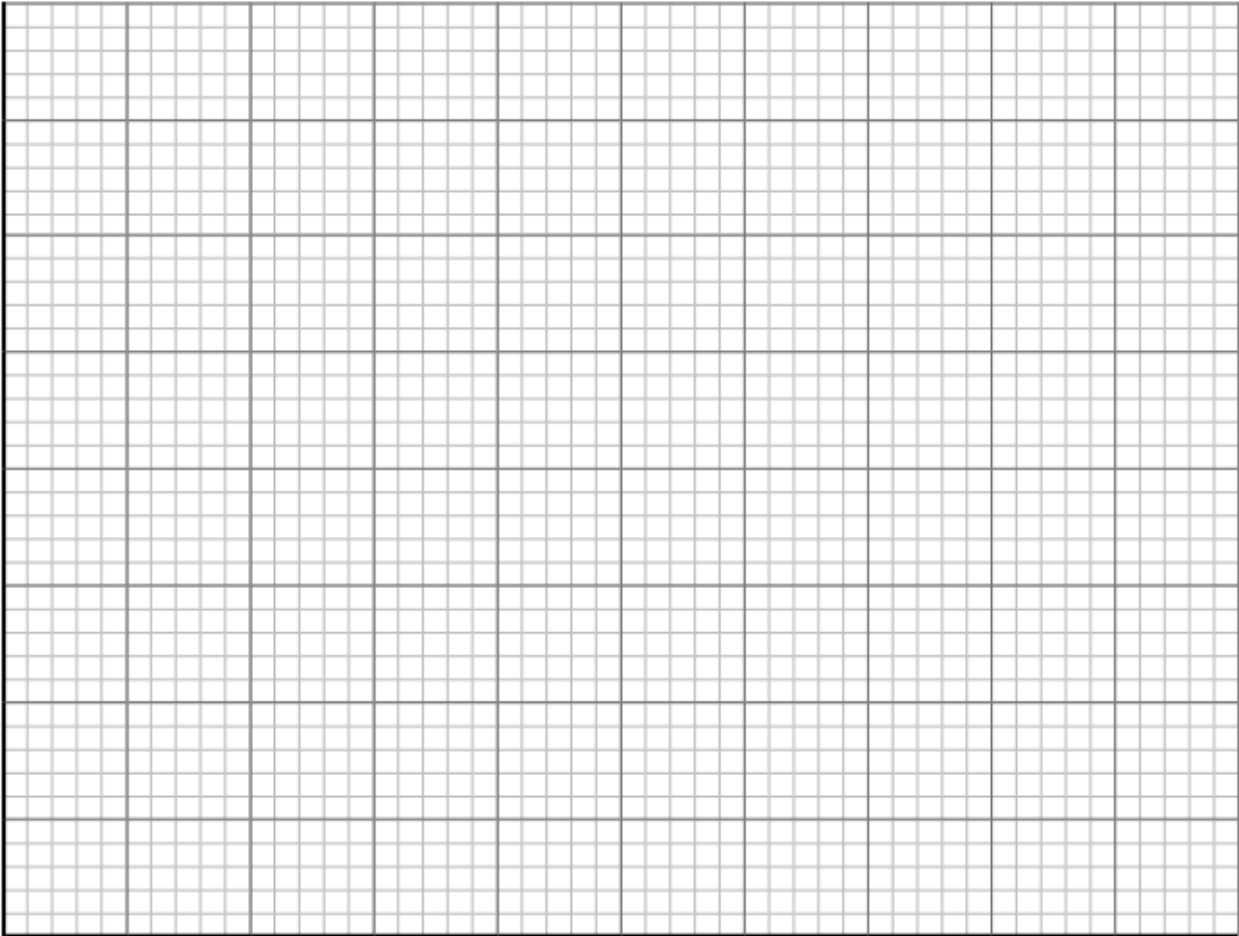
Therefore no sound is heard in space - it is completely silent.

Speed of Sound

The table below shows the speeds at which sound waves travel through different materials.

Medium	State	Speed of Sound (m/s)
air	gas	343
helium	gas	1007
water	liquid	1498
seawater	liquid	1531
glycerine	liquid	1904
aluminium	solid	3100
steel	solid	3235
glass	solid	3280

Use the information in the table to draw a bar chart showing the speeds at which sound waves travel through different materials. You may like to use a key to show the state of each material.



· Bar chart drawn from the table: one bar per material with height = the speed of sound in that material (m/s). Label both axes with units, add a title, and use a key (colours) to show whether each material is a solid, liquid or gas. Bars should be tallest for solids, medium for liquids and shortest for gases.

Unit 9.5 – 9.6 - Sound

Use the data table and the bar chart you have drawn to answer the following questions.

1. State the speed of sound in air.

343 m/s

2. Describe the pattern shown in the data table and bar chart.

The speed of sound is highest in solids and lowest in gases - it decreases from solid to liquid to gas.

For example air 343 m/s, water about 1500 m/s and steel about 5900 m/s, so solids are many times faster than gases.

3. Use your knowledge of the arrangement of particles in the different states of matter to explain this pattern.

In a solid the particles are packed closely together and strongly bonded to each other.

A vibration is therefore passed straight on from particle to particle very quickly, so sound travels fast.

In a gas the particles are far apart, so they only collide occasionally and the vibration takes much longer to be passed along.

This means the closer the particles are, the faster the sound wave travels.

Custard is an example of a non-Newtonian fluid. This means it can act like a liquid or a solid depending on the pressure put on its surface.

4. Suggest a value for the speed of sound through custard. Give a reason for your answer.

1500 m/s

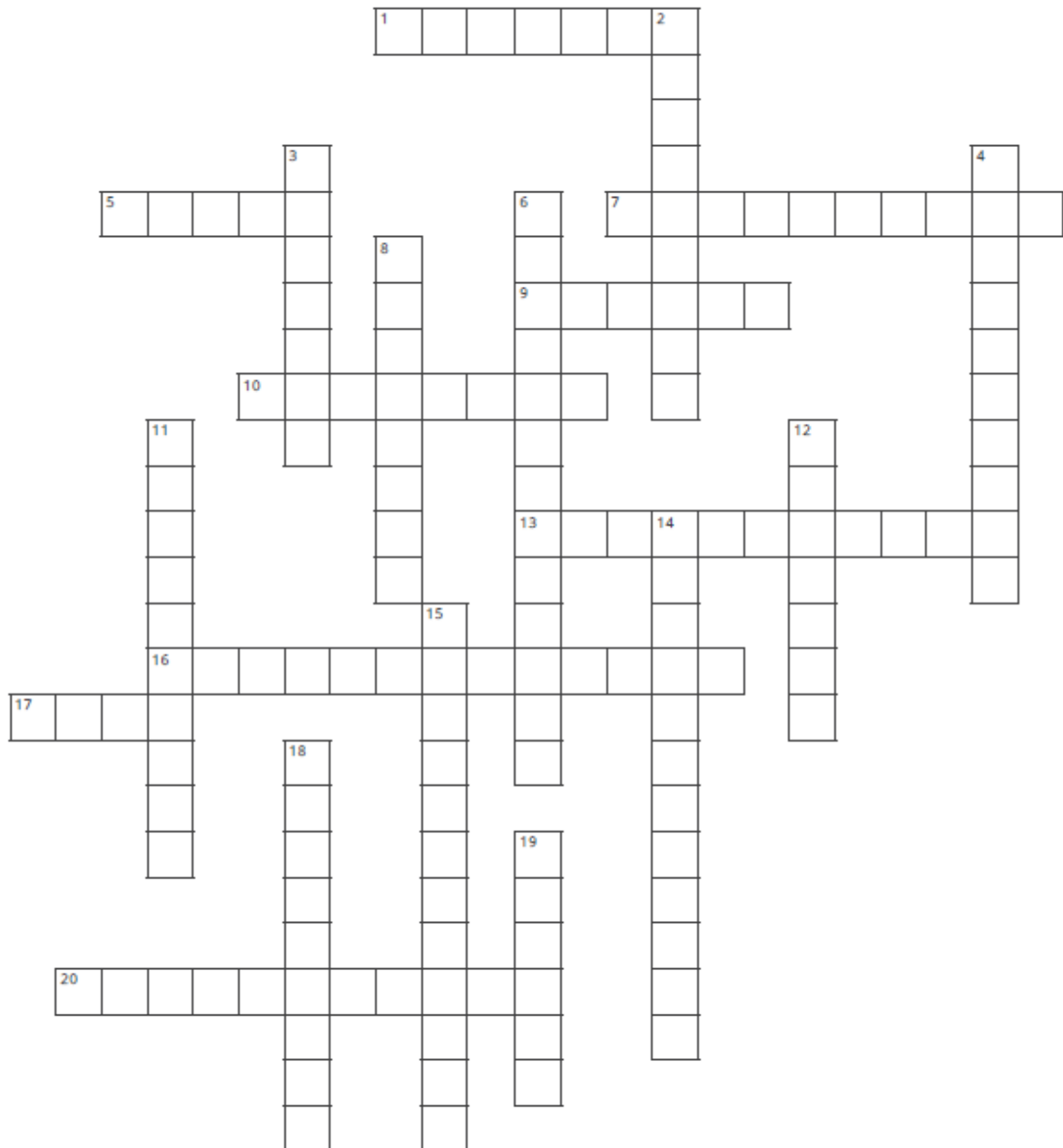
Custard is a non-Newtonian fluid - it behaves like a thick liquid when left alone but acts like a solid when hit hard.

Its particles are much closer together than air, but not as rigidly fixed as a solid, so a value around 1500 m/s

(between a liquid and a solid) is sensible of sound in air (343 m/s) because the particles are packed much closer together.

Sound and Hearing Crossword

Solve the clues about sound and hearing to complete the crossword. The number at the end of each clue tells you how many letters there are in the answer.



Across

1. The fluid-filled organ of the inner ear which creates electrical impulses. (7)
5. The external part of the ear which funnels sound waves into the ear canal. (5)
7. A device used by singers to convert the vibrations from sound waves into electrical signals. (10)
9. An absence of matter that sound waves cannot travel through. (6)
10. The nerve which transmits electrical impulses from the cochlea to the brain. (8)
13. An area on a longitudinal wave where the particles are further apart than average. (11)
16. The ability of waves to add together or cancel each other out when they travel together or cross a point. (13)
17. A reflected wave. (4)
20. An area on a longitudinal wave where the particles are closer together than average. (11)

Down

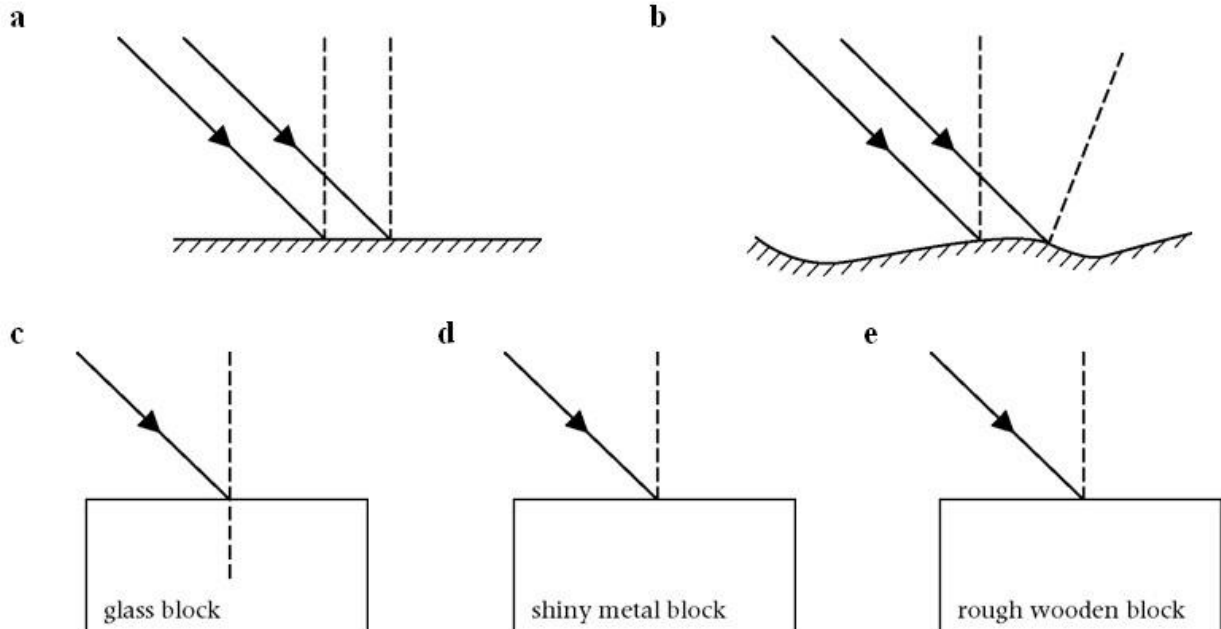
2. The distance between the resting position of a wave and the furthest point from it. (9)
3. A thin membrane inside the ear that vibrates when energy is transferred to it from vibrating particles. (7)
4. Sound waves with frequencies below the lower limit of human hearing. (10)
6. The superposition of several echoes. (13)
8. The collective term for the three tiny bones of the middle ear. (8)
11. Sound waves with frequencies above the upper limit of human hearing. (10)
12. The unit for measuring the loudness of sound. (7)
14. Animals like bats and dolphins use this to locate prey and obstacles. (12)
15. Sound waves are an example of this type of wave. (12)
18. The number of waves passing a point each second. (9)
19. The value, in hertz, of the frequency at the lower end of the human audible range. (6)

Time for a quiz [10. Sound Waves-Quiz.ppt](#)

<https://docs.google.com/presentation/d/1F5pL1Z0IXmBWyaEnkGpfb-UGmBZ0J2mO/edit#slide=id.p1>

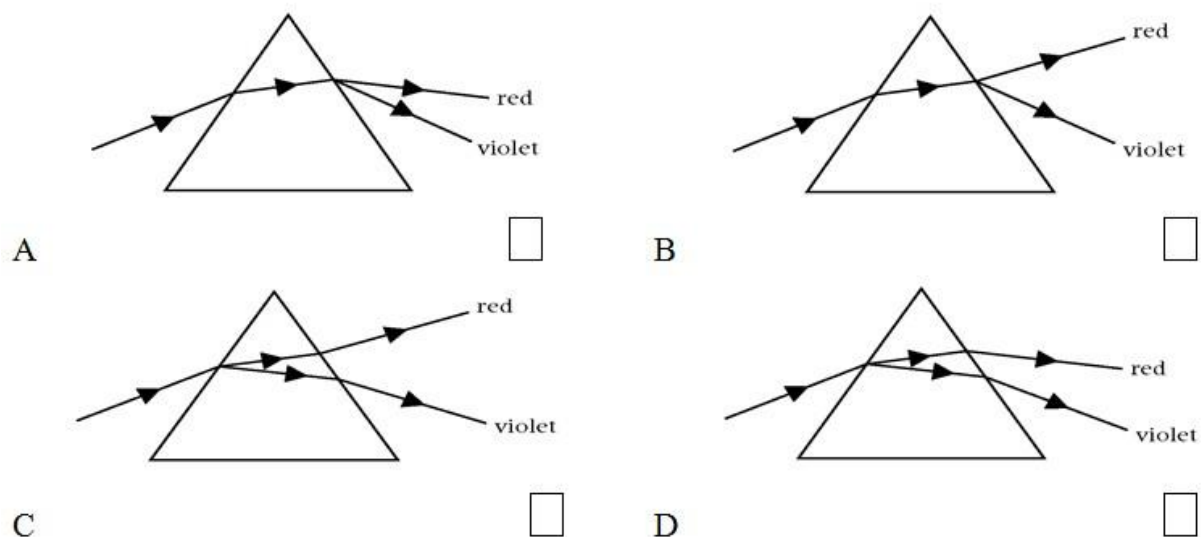
Quick Quiz

1 Complete the rays on the drawings below.



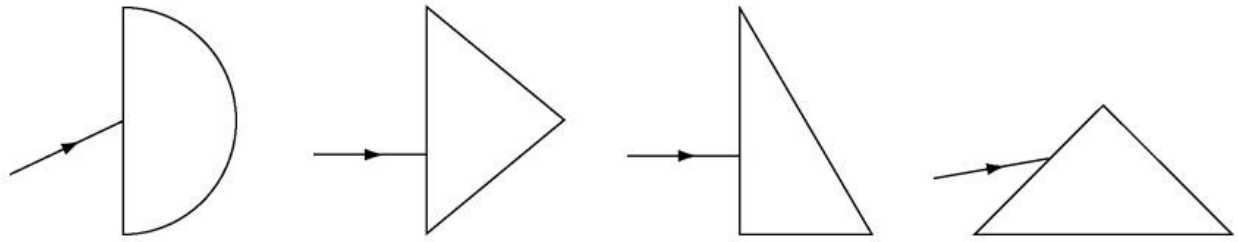
· Complete the ray diagrams: draw the normal (dashed line at 90 degrees to the mirror), then the reflected ray on the other side of the normal with the angle of reflection equal to the angle of incidence, both measured from the normal.

2 Put a tick by the correct diagram.



· Tick the diagram that correctly shows the law of refraction - the one where the angle of incidence equals the angle of refraction, both measured from the normal, and with the incident ray, refracted ray and normal all in the same plane.

3. Complete the following ray diagrams



Extension: Find out about the position of visible light in the **electromagnetic spectrum**. How is visible light different to UV, IR, Gamma or X rays?

· Complete the ray diagrams: light entering the glass block slows down and bends towards the normal at the first interface, then speeds up and bends away from the normal as it leaves. Draw the normal at 90 degrees to each surface and continue the ray straight to the eye.

Summary Sheets

Light

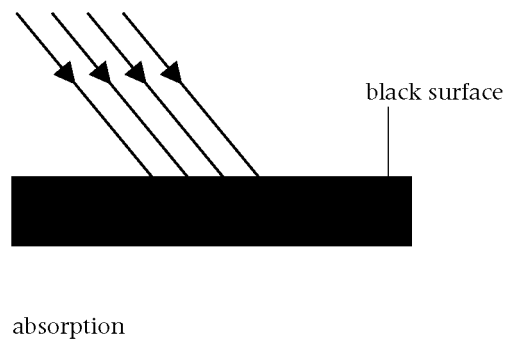
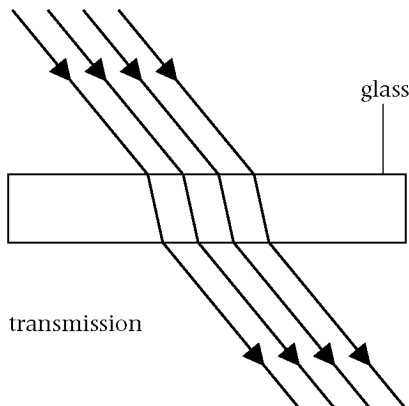
Objects which create light are **luminous sources**. Objects are luminous when they have a high temperature, like the surface of the sun.

Light travels in *straight* lines.

Light waves travel through **transparent** objects but not through **opaque** objects. **Shadows** are made because light cannot travel through opaque objects. **Translucent** objects show a glow of light through them.

Transmission and absorption

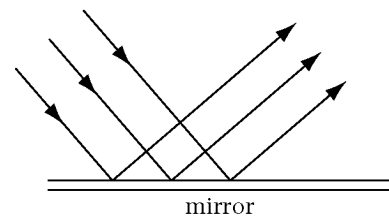
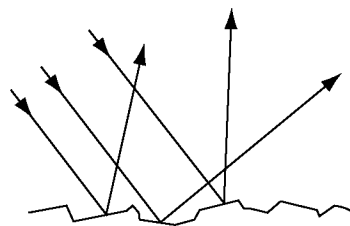
Transparent materials let light pass straight through. We say they **transmit** light. Opaque surfaces can **absorb** light. Black surfaces absorb light very well and reflect very little. This is why they look so dark.



Reflection

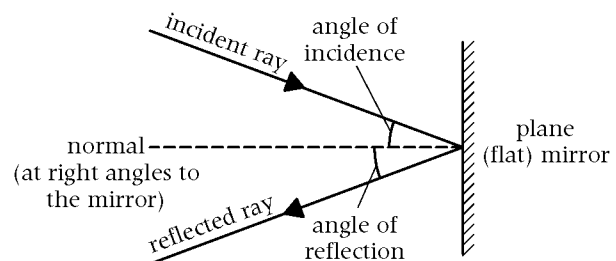
Light rays are **scattered** by rough surfaces, and a **reflection** cannot be seen.

A **plane** mirror is a flat mirror. Light is reflected evenly by a plane mirror.



The **angle of incidence** is equal to the **angle of reflection**.

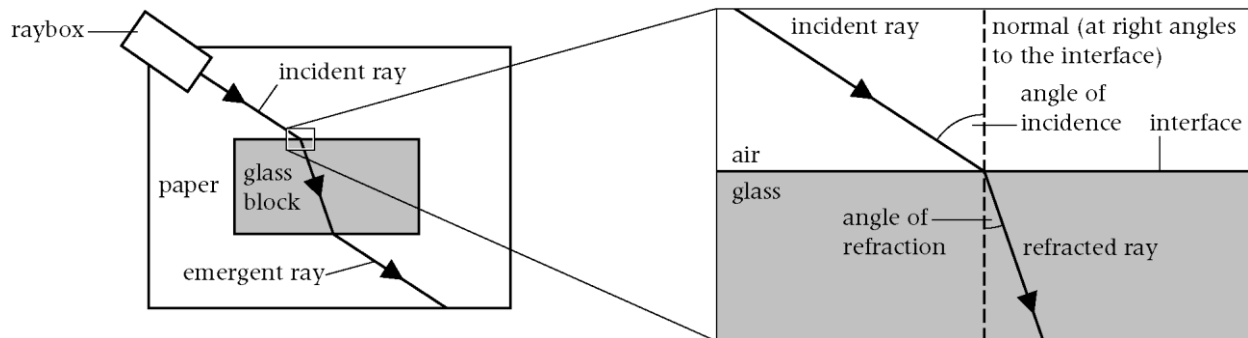
When light shines on to an object viewed in a mirror, the rays are **reflected** into the eye. They seem to come from a position behind the mirror. The **image** is the **same size** as the object and the **same distance** from the mirror. In the image left is right and right becomes left.



Refraction

When light hits something **transparent** it changes direction. This is called **refraction**.

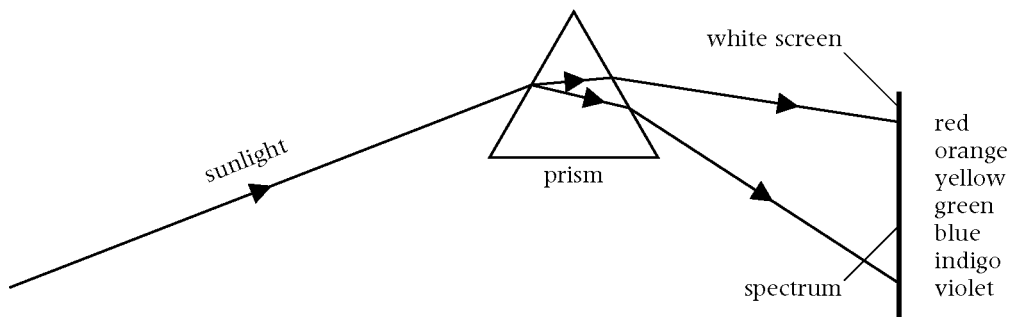
Refraction takes place at the **interface** between two substances. When light is transmitted through glass it slows down and bends towards the **normal**. When it travels back out it speeds up again and bends away from the normal.



Colour

White light is a mixture of colours. White light can be split up using a **prism** to give a **spectrum** of seven visible colours (red, orange, yellow, green, blue, indigo, violet).

The splitting of colour into a spectrum is called **dispersion**.



Different colours can be made by mixing light of the three **primary colours** (red, green and blue).

Coloured light can be made using a **filter**. A red filter lets red light through, but **absorbs** all the other colours.

We are able to see colours because objects do not reflect all the colours in light:

White objects reflect all the colours.

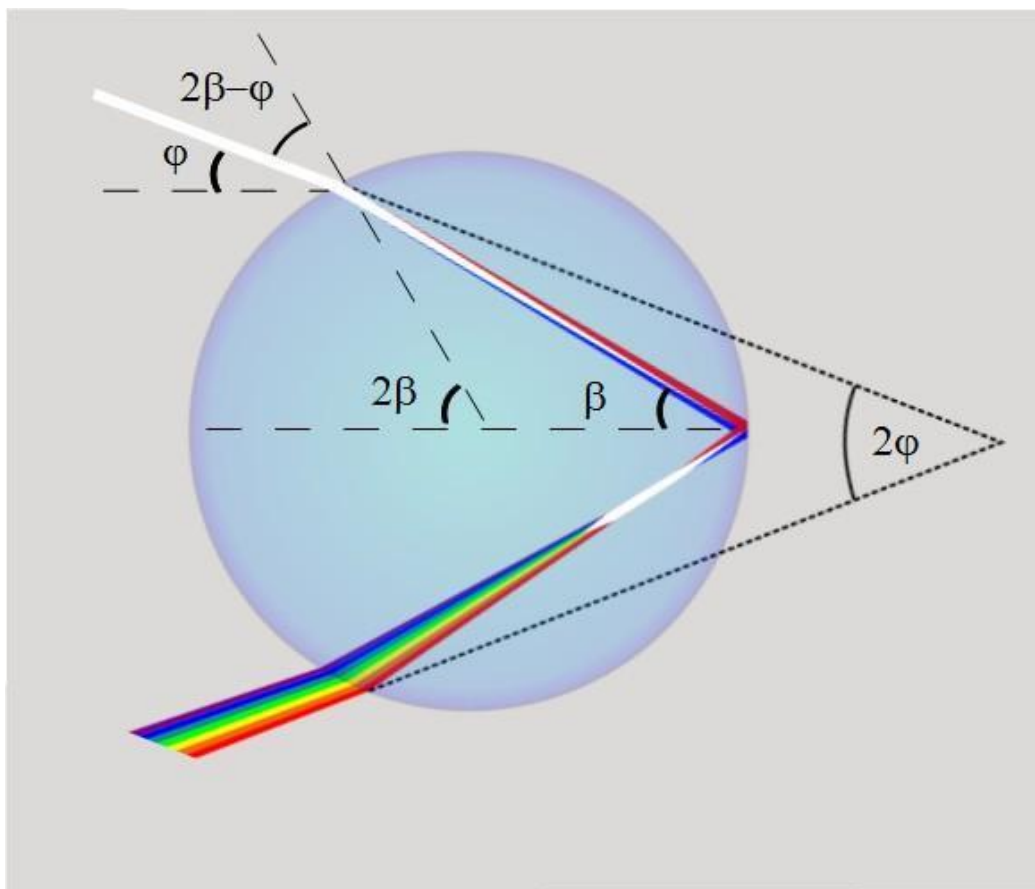
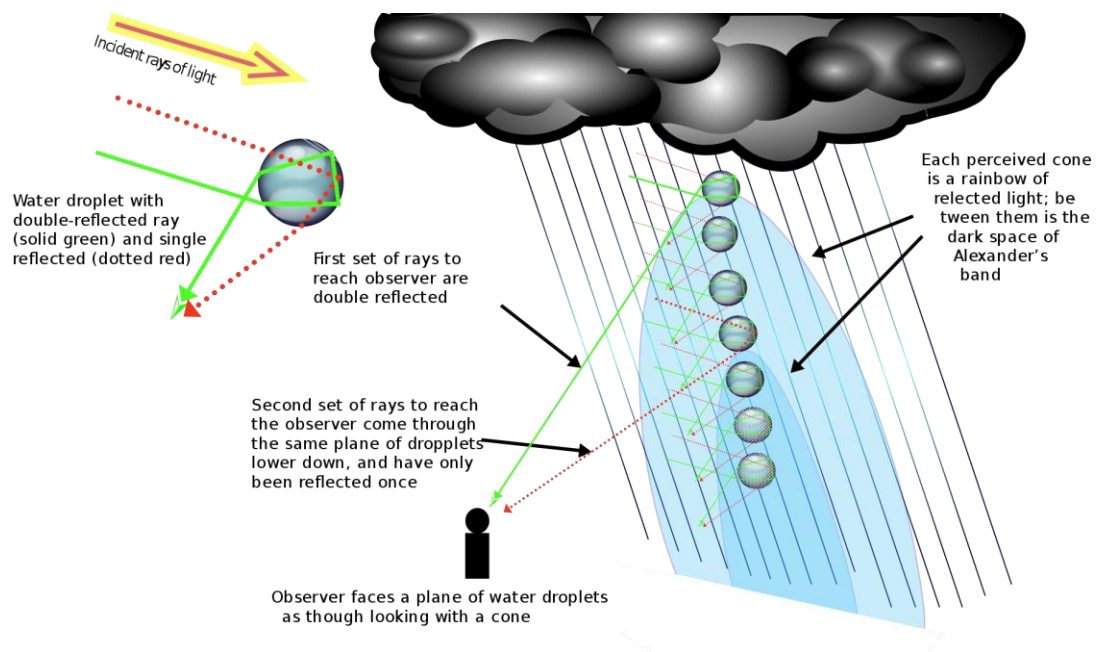
A red object only reflects red and all other colours are absorbed.

This idea applies to all colours except black.

Black objects absorb all colours.

Rainbow

A **rainbow** is produced when water droplets in the air **refract** sunlight.



Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p9: The left-hand image is correct. The image is the right way up but reversed left to right (laterally inverted), not upside down.

p28: Diagram: light from the object travels in straight lines through the pinhole and forms an image on the screen at the back of the box. The image is inverted because the rays cross over at the pinhole.

p33: Labelling: the hole in the centre of the front face is the 'small hole'; the ray drawn from the top of the candle flame is the 'light ray from the top of the candle flame' and the one from the bottom of the flame is the 'light ray from the bottom of the candle flame'; the picture on the screen is the 'upside down image'; the caption is 'Light from one part of the flame can only reach one place on the screen.'

p44: Plot power (y-axis, D) against focal length (x-axis, cm). The line falls steeply and curves - power is inversely proportional to focal length (power = $1/f$).

p46: A refracting telescope uses two convex lenses: a large objective lens with a long focal length collects the light and forms a real, inverted image, and a smaller eyepiece lens with a short focal length magnifies that image for the eye.

p61: Delete (a): a polarizing filter does not twist the waves - it blocks part of the vibrations and only lets through those in a specific plane.

p71: Stars 5-10: 5 red giant, 6 star moving away (redshifted), 7 star with a large planet, 8 binary pair, 9 Cepheid variable, 10 nebula. Left cell of each row = colour, right cell = class.

p93: Sound is a vibration which travels through a medium (e.g. air, water, a solid) and transfers energy from one place to another; it is made when something vibrates. The tree still makes sound waves because it pushes the air particles around it, but there is nobody there to hear it (hearing needs an ear).

p99: Particle models - Solid: particles drawn tightly packed in a regular pattern, touching. Liquid: particles close together but randomly arranged, with small gaps, able to slide past each other. Gas: particles far apart and randomly arranged, moving quickly in all directions.